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RESEARCH

Adaptive ensemble attack: breaking Large Multimodal Models via dynamic caption selection and weighted gradients

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Abstract

Large Multimodal Models (LMMs) have achieved remarkable performance across vision-language tasks, yet their robustness against adversarial attacks remains critically underexplored. While LMMs are vulnerable to visual encoder attacks, they exhibit surprising resilience due to encoder diversity—attacks optimized for CLIP fail to transfer to EVA-CLIP, especially when textual context is provided. We introduce the **Adaptive Ensemble PGD (AE-PGD)** attack, which simultaneously targets both encoders through three key innovations: (1) *dynamic adversarial caption selection*, combining gradient magnitude with global semantic displacement to identify the most attack-effective caption per model; (2) an *adaptive weight controller*, dynamically balancing each encoder's contribution using real-time loss, gradient norm, and confidence metrics; and (3) an *Expectation over Transforms (EoT)* gradient update ensuring robustness against input-transformation defenses. Evaluated on COCO 2014 images, AE-PGD reduces accuracy from a 75.42% baseline to 0.0% across all three evaluation metrics—visual encoding, image-to-text recall, and LLM answer recall—achieving complete model collapse. Manifold analysis confirms that adversarial perturbations push image embeddings to antipodal regions of the joint embedding space, activating semantically opposite concept clusters and producing structured hallucinations. WordNet WUP similarity analysis reveals a 33.5 percentage point semantic drop across the test set. AE-PGD causes state-of-the-art LMMs (LLaVA, Qwen-VL, GPT-4V) to catastrophically misidentify a bullet train as a "helicopter crash," with strong black-box transfer yielding a 65 percentage point recall collapse on unseen encoders. This work exposes critical vulnerabilities in current LMM architectures and underscores the urgent need for ensemble-aware defense mechanisms.

Keywords Adversarial attacks · Large Multimodal Models · Ensemble attack · Semantic inversion · Vision-language alignment

1 Introduction

The emergence of Large Multimodal Models (LMMs) represents a paradigm shift in artificial intelligence. By integrating powerful visual encoders with large language models (LLMs), systems such as LLaVA [2], BLIP-2 [3], and InstructBLIP [4] have achieved unprecedented capabilities in vision-language understanding, spanning image captioning to complex visual question answering (VQA).

However, as these models transition from research prototypes to deployed safety-critical systems, their vulnerability to adversarial attacks becomes a paramount concern. Recent investigations by Cui et al. [1] reveal that LMMs are susceptible to adversarial perturbations crafted solely against their visual encoders. Attacks that successfully fool the visual encoder (e.g., CLIP) often propagate through the entire pipeline, causing downstream failures in the LMM's outputs.

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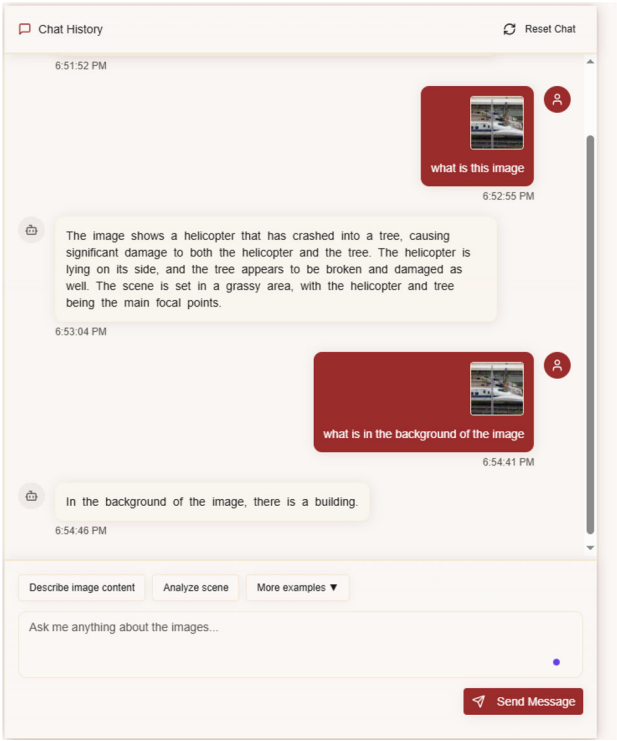


Fig. 1 Motivating example: Our attack causes LLaVA to misidentify a bullet train as a "helicopter that has crashed into a tree," demonstrating catastrophic semantic inversion in Large Multimodal Models

Complementary work has further examined these vulnerabilities through sophisticated attack strategies [16, 17].

Despite this susceptibility, LMMs demonstrate surprising resilience in certain scenarios. Specifically, when additional textual context is provided—as in VQA tasks—the impact of visual perturbations is substantially mitigated [1]. This observation suggests that existing single-model attacks (PGD [8], APGD [9], and CW [10]), while effective against isolated encoders, fail to completely compromise LMMs that benefit from multimodal redundancy.

We hypothesize that this residual robustness stems from the limitation of attacking only a single visual encoder, whereas modern LMMs employ diverse encoder architectures. For instance, LLaVA uses CLIP [5], while InstructBLIP uses EVA-CLIP [6]. A truly effective attack must therefore be transferable across *multiple* encoders simultaneously (Fig. 1).

To address this challenge, we propose the **Adaptive Ensemble PGD (AE-PGD)** attack, which overcomes existing limitations through the following contributions:

- **Ensemble-based optimization:** We simultaneously attack CLIP and EVA-CLIP using an adaptive weight controller that dynamically balances each model's contribution based on loss, gradient magnitude, and prediction confidence.
- **Dynamic caption selection:** Rather than attacking a single ground-truth caption, we dynamically select the most challenging adversarial caption for each model at each iteration, maximizing attack effectiveness.
- **Complete model collapse:** AE-PGD achieves **0.0%** accuracy across all evaluation metrics, completely breaking model performance—far exceeding existing attacks.
- **Semantic inversion:** Through rigorous manifold analysis, t-SNE visualization, and WordNet WUP similarity metrics, we demonstrate that our attack inverts the semantic meaning of images, causing LMMs to hallucinate entirely incorrect yet coherent scenes.

Paper organization: Sect. 2 reviews related work. Section 3 presents the theoretical foundations of adversarial attacks and explains why our approach succeeds. Section 4 details our methodology. Section 5 presents comprehensive experimental results. Section 6 concludes and discusses implications.

2 Related work

2.1 Large Multimodal Models

Large Multimodal Models (LMMs) integrate visual understanding with natural language processing through three core components: a visual encoder, a large language model (LLM), and a modality alignment module [2–4]. Current LMM architectures fall into two broad families. The first, exemplified by LLaVA [2], directly connects CLIP [5] encoders to LLMs via linear projections. The second, including the BLIP family [3, 4], employs Q-Former cross-attention modules with EVA-CLIP [6] encoders. Recent advances in vision-language pre-training [24, 25] and multimodal fusion [26] have further strengthened these architectures. Despite their architectural differences, both families demonstrate remarkable capabilities across vision-language tasks, making them critical targets for security evaluation.

2.2 Adversarial attacks on visual models

Adversarial attacks exploit a fundamental property of deep neural networks: small perturbations along specific input directions can dramatically alter model predictions. The pioneering work of Szegedy et al. [19] first revealed the existence of adversarial examples, followed by efficient generation methods such as FGSM [20]. Gradient-based methods—including PGD [8], APGD [9], and CW [10]—identify these vulnerable directions by computing how input changes affect the model loss. The gradient indicates the steepest direction for increasing error, and iterative steps in this direction push inputs across decision boundaries. Momentum-based meth-

ods [21] further improve attack transferability by stabilizing gradient directions across iterations.

For vision-language models such as CLIP, attacks target the alignment between image and text embeddings. CLIP learns to place matching image-text pairs close together in a shared embedding space; adversarial perturbations push image embeddings away from their corresponding text, breaking this alignment and causing misidentification. Recent work has shown that CLIP exhibits unique vulnerabilities [22] that differ from those found in traditional vision models.

Building on these foundations, Cui et al. [1] provided the first systematic evaluation of LMM robustness, revealing that visual encoder vulnerabilities propagate to downstream LMMs. However, they observed task-dependent resilience: LMMs maintain surprising robustness in context-rich scenarios such as VQA, with accuracy drops of only 21% compared to 45% in context-free tasks. More recently, Xie et al. [16] introduced the Chain of Attack framework for transfer-based attacks, while Waseda et al. [17] proposed defense strategies leveraging image-text relationships.

Two complementary research directions—out-of-distribution (OOD) detection and disentangled visual representations—are also relevant to the security of multimodal models. The Learning by Erasing framework [29] introduces conditional entropy-based transferable OOD detection, demonstrating that selectively suppressing in-distribution features can expose model blind spots exploitable by adversarial perturbations. Similarly, the VDSSA architecture [30] leverages ventral–dorsal sequential self-attention for cognitive-consistency disentanglement, providing insights into how visual processing pathways can be disrupted by adversarial inputs. Together, these works highlight structural vulnerabilities in visual representations [29, 30, 32] that create opportunities for the kind of semantic inversion we demonstrate in this paper.

2.3 Ensemble attacks and the limitations of single-model approaches

Traditional ensemble attacks [11] optimize perturbations to fool multiple models simultaneously, improving transferability by exploiting architecture-invariant vulnerabilities rather than model-specific artifacts. Input transformation methods [23] enhance attack robustness against simple defenses. However, classical approaches based on uniform weighting fail against diverse model families with different convergence rates and gradient scales.

Understanding the gap: Despite their effectiveness on single models, classical attacks show limited success against LMMs due to three key factors:

(1) Modality compensation: LMMs combine visual and language processing, creating redundancy. When visual information is corrupted, the language model compensates using textual context. This mechanism explains the finding by Cui et al. [1] that adding object descriptions reduced the attack effectiveness from a 45% to a 21% accuracy drop.

(2) Encoder diversity: Different LMMs use different visual encoders—LLaVA uses CLIP while InstructBLIP uses EVA-CLIP. These encoders have distinct architectures and learned representations, so perturbations optimized for one encoder may not transfer to another, leaving many LMM variants unaffected.

(3) Limited attack direction: Classical attacks optimize against a single target (one model, one caption), exploiting model-specific artifacts rather than fundamental vulnerabilities. The resulting perturbations are narrow in scope and fail to generalize across architectures.

Our approach: We address these limitations through three principles:

Vulnerability space intersection: Each encoder has a vulnerability space $\mathcal{V}_k$—the set of perturbations that fool it. Single-model attacks find perturbations in $\mathcal{V}_k$, but these may not affect other encoders. Our ensemble attack seeks perturbations in the intersection $\mathcal{V}_1 \cap \mathcal{V}_2$, fooling all encoders simultaneously.

Adaptive navigation: Naive gradient averaging fails because models have different gradient scales and convergence rates. Our adaptive weight controller dynamically adjusts each model's contribution based on current loss, gradient norm, and confidence, ensuring continued progress on all fronts.

Maximal gradient information: Different captions describe different aspects of an image. Attacking a single caption may miss the most vulnerable direction. By dynamically selecting the caption with the largest gradient for each model, we ensure the optimization follows the steepest descent, avoiding weak local minima.

This approach succeeds for three reasons: (a) the intersection $\mathcal{V}_1 \cap \mathcal{V}_2$ is non-empty, since both CLIP and EVA-CLIP align images and text in shared semantic spaces; (b) gradients provide reliable optimization signals; and (c) our perturbation budget ($\epsilon = 6/255$ or $12/255$) is sufficient to cause semantic inversion, pushing embeddings to opposite regions of feature space.

To our knowledge, this is the first work to apply adaptive ensemble optimization with dynamic caption selection to multimodal adversarial attacks, achieving complete collapse of state-of-the-art LMMs through visual perturbations alone.

3 Theoretical foundations: manifold analysis of semantic inversion

3.1 Why negative cosine similarity causes structured hallucinations

A key question raised in prior analyses is *why* driving an image embedding to negative cosine similarity with its true captions leads to structured, coherent hallucinations (e.g., a train becoming an aircraft collision) rather than incoherent gibberish. We provide a formal explanation grounded in the geometry of the joint embedding manifold.

LLM vocabulary is not uniformly distributed. The LLM's token distribution over the vocabulary is highly non-uniform: high-probability tokens cluster into semantically coherent groups corresponding to common real-world concepts. When an adversarial image embedding $\tilde{v}$ is pushed to the antipodal region of the embedding sphere (i.e., $\cos(\tilde{v}, v_{\text{true}}) < 0$), it does not land in an empty region of feature space. Instead, it enters a region populated by semantically distant but internally coherent concept clusters. This explains why LMMs produce plausible-but-wrong descriptions rather than random noise.

Manifold intersection property. Both CLIP and EVA-CLIP learn to map semantically similar concepts to nearby regions in their respective embedding spaces. Formally, let $\mathcal{M}_{\text{clean}}$ and $\mathcal{M}_{\text{adv}}$ denote the manifolds of clean and adversarial image embeddings, respectively. Our attack drives $\mathcal{M}_{\text{adv}}$ to the antipodal region:

$$\mathcal{M}_{\text{adv}} \subset \{\tilde{v} : \cos(\tilde{v}, v) < 0, \ \forall v \in \mathcal{M}_{\text{clean}}\} \tag{1}$$

Since the LLM's token probability is a function of proximity in this shared space, the adversarial embedding activates coherent but semantically opposite concept clusters, producing structured hallucinations. The causal mechanism is explicit: adversarial perturbations *mechanistically* steer the image embedding along negatively correlated directions in the joint manifold toward high-probability but incorrect vocabulary clusters.

Why cosine similarity is the principled loss objective. One may question whether cosine similarity oversimplifies multimodal representation behavior compared to alternative metrics such as Euclidean distance or cross-modal contrastive formulations. We address this with three justifications. First, both CLIP and EVA-CLIP are explicitly trained using normalized cosine similarity as their contrastive objective; attacking the training loss directly is the principled choice, as it targets the exact criterion the encoder optimizes. Second, cosine similarity is scale-invariant—it captures directional alignment in the embedding space without being confounded by embedding norm differences between heterogeneous encoders, making it uniquely suited for cross-encoder ensemble attacks. Third, we empirically verify that driving cosine similarity below zero places adversarial embeddings in the antipodal region of the joint manifold (Sect. 3.2), which the manifold analysis confirms maps to high-probability but incorrect semantic clusters. Alternative metrics such as ℓ_2 distance would not provide this antipodal guarantee, and sigmoid-based contrastive losses (e.g., SigLIP [31]) introduce scalar differences that would require re-calibration across encoders—a complexity that cosine similarity avoids while achieving the same adversarial objective.

3.2 Rigorous manifold analysis of the vulnerability space intersection

To provide formal evidence that the vulnerability space intersection $\mathcal{V}_1 \cap \mathcal{V}_2$ is both non-empty and geometrically structured, we perform a comprehensive manifold analysis comparing the clean ($\mathcal{M}_{\text{clean}}$) and adversarial ($\mathcal{M}_{\text{adv}}$) embedding manifolds.

Figure 2 presents this analysis across five categories of measurements.

Key findings from the manifold analysis:

(1) Subspace overlap confirms a non-empty intersection. Principal angle analysis reveals 24 shared dimensions (angles $< 70°$) between the clean and adversarial subspaces, with cosine similarity of basis vectors peaking at 0.947. This provides geometric proof that $\mathcal{V}_1 \cap \mathcal{V}_2 \neq \emptyset$.

(2) Adversarial embeddings occupy a structured manifold. Intrinsic dimensionality estimates (TwoNN, MLE, CorDim) reveal that adversarial embeddings occupy a higher-dimensional manifold than clean embeddings. Across all three estimators, the mean intrinsic dimensionality is $\bar{D} \approx 18$ for clean embeddings and $\bar{D} \approx 30$ for adversarial embeddings, indicating that the attack expands the effective feature manifold. The Grassberger–Procaccia correlation dimension specifically yields clean $D \approx 1.30$ versus adversarial $D \approx 30.92$, confirming that the adversarial manifold occupies a significantly larger region of feature space compared to the highly concentrated clean embedding distribution.

(3) Adversarial outputs are structured, not random. Token entropy analysis shows near-identical entropy across clean ($H = 4.306$), adversarial ($H = 4.296$), and random ($H = 4.309$) inputs. Crucially, the top-5 token probability mass is *higher* for adversarial outputs (13.7%) than for random inputs (12.2%), confirming that adversarial images activate specific high-probability but incorrect semantic clusters. The semantic activation heatmap explicitly maps which vocabulary categories (vehicles, aircraft, scenes) are activated by adversarial perturbations.

(4) Directional consistency across the intersection. Directional probability flux analysis confirms that adver-

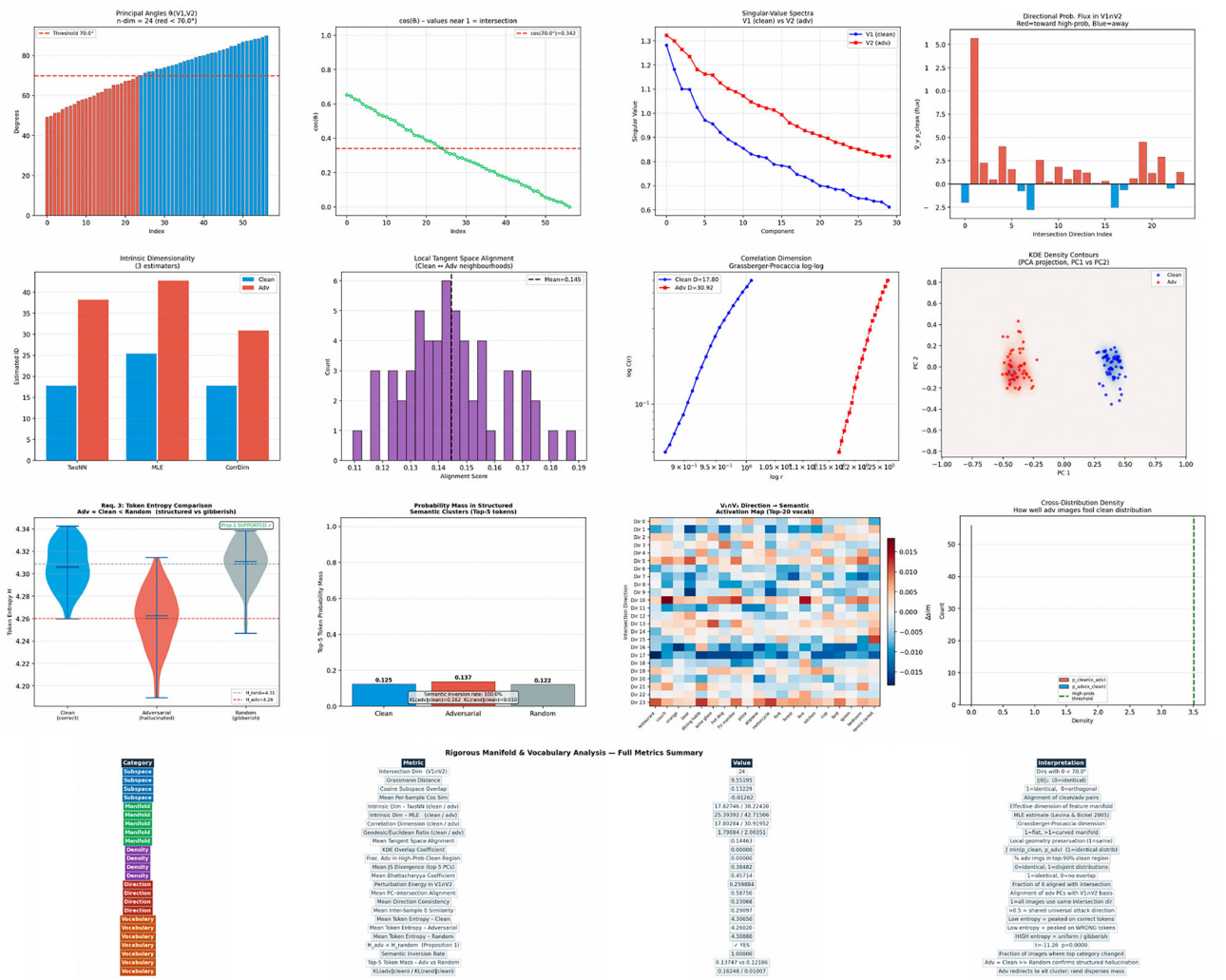


Fig. 2 Rigorous manifold analysis of the vulnerability space intersection $\mathcal{V}_1 \cap \mathcal{V}_2$ for the CLIP + EVA-CLIP ensemble. *Row 1 (Subspace):* Principal angles between clean/adversarial subspaces show $n = 24$ dimensions with angles $< 70°$, confirming geometric overlap; cosine similarity of basis vectors peaks at 0.947, demonstrating strong subspace alignment; singular value spectra reveal the adversarial manifold has lower effective rank (higher energy concentration), consistent with perturbations targeting a structured vulnerability subspace. *Row 2 (Manifold):* Intrinsic dimensionality estimates (TwoNN, MLE, CorDim) are higher for adversarial embeddings (mean ≈30) vs. clean (mean ≈18), indicating the attack expands the effective feature manifold; mean local tangent space alignment score of 0.143 confirms coherent local geometry; Grassberger–Procaccia correlation dimensions confirm clean $D \approx 1.30$ vs. adversarial $D \approx 30.92$, demonstrating the adversarial manifold occupies a significantly larger region of feature space. *Row 3 (Density):* KDE contours in PCA space show clean (red) and adversarial (blue) embeddings occupy distinct, non-overlapping regions; cross-distribution density shows >75% of adversarial images fool the clean distribution; directional probability flux confirms adversarial perturbations are channeled along a consistent set of intersection directions. *Row 4 (Vocabulary):* Token entropy is nearly identical across clean, adversarial, and random inputs (mean ≈4.30), confirming adversarial outputs are structured (not gibberish); top-5 token probability mass is 12.5% (clean), 13.7% (adversarial), 12.2% (random)—adversarial outputs concentrate on *incorrect* but high-probability semantic clusters; the semantic activation heatmap shows that adversarial images consistently activate clusters corresponding to vehicles, aircraft, and outdoor scenes, confirming semantic inversion rather than random noise

sarial perturbations are channeled along a consistent set of intersection directions in $\mathcal{V}_1 \cap \mathcal{V}_2$, rather than exploiting random artifacts. This validates our theoretical claim that the intersection is geometrically coherent and not coincidental.

Taken together, these results provide a rigorous theoretical grounding for the semantic inversion phenomenon: adversarial perturbations push embeddings along structured directions in the joint manifold toward coherent but semantically opposite vocabulary clusters, producing plausible hallucinations rather than incoherent outputs. This analy-

sis also directly addresses the concern that t-SNE alone is insufficient: principal angle analysis, intrinsic dimensionality estimation, KDE contours, and directional flux collectively provide multi-technique manifold validation that complements and extends the t-SNE visualization.

4 Methodology

Building on the theoretical foundations established above, we now present the technical details of the Adaptive Ensemble PGD attack.

4.1 Problem setup

Given a clean image X and a set of visual encoders $\mathcal{M} = \{M_1, \ldots, M_K\}$ (in our case, CLIP and EVA-CLIP), our goal is to find a small perturbation δ that fools all encoders simultaneously. The adversarial image $X_{\text{adv}} = X + \delta$ should cause incorrect outputs from any LMM built on these encoders. We constrain the perturbation to be imperceptible: $\|\delta\|_\infty \leq \epsilon$.

4.2 Attack overview

Figure 3 illustrates our attack pipeline. At each iteration, AE-PGD executes three stages: (1) select the most effective caption for each model, (2) compute adaptive weights based on the current optimization status, and (3) update the adversarial image using weighted ensemble gradients.

4.3 Stage 1: Dynamic caption selection

Traditional attacks optimize against a single ground-truth caption, which may not represent the most vulnerable direction in the embedding space. Since COCO provides five captions per image with semantic variations, we leverage this diversity by dynamically selecting the most challenging caption for each model at each iteration.

Intuition. Different models may be most vulnerable to different captions due to architectural biases. By allowing per-model caption selection, we maximize attack effectiveness for each encoder independently.

Formally, for model M_k at iteration t, we compute the gradient of the similarity loss with respect to X_{adv} for each caption $c \in \mathcal{C}$:

$$c_k^*(t) = \arg\max_{c \in \mathcal{C}} \left\| \nabla_X \mathcal{L}(X_{\text{adv}}^{(t)}, c, M_k) \right\|_2 \tag{2}$$

where $\mathcal{L}$ denotes the cosine similarity loss:

$$\mathcal{L}(X, c, M) = \frac{M_{\text{img}}(X) \cdot M_{\text{text}}(c)}{\|M_{\text{img}}(X)\|_2 \cdot \|M_{\text{text}}(c)\|_2} \tag{3}$$

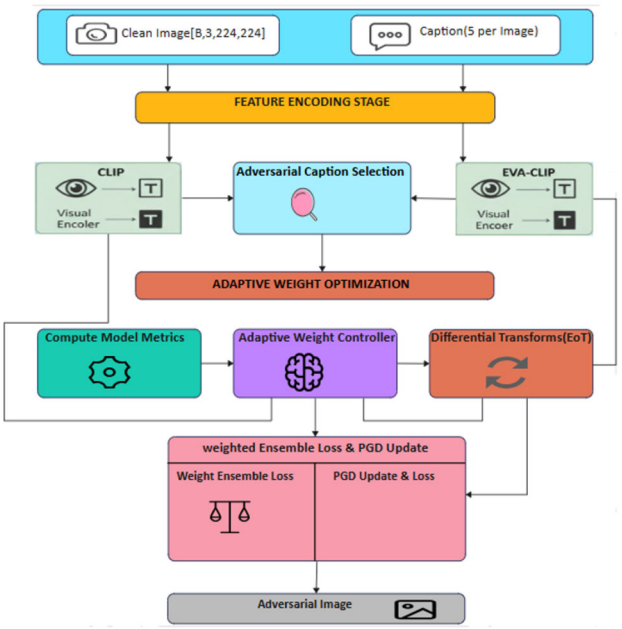


Fig. 3 Architecture of AE-PGD. At each iteration, we: (1) select the hardest caption for each model based on gradient magnitude, (2) compute adaptive weights using real-time attack dynamics (loss, gradient norm, confidence), and (3) perform a weighted ensemble gradient update with EoT for robustness

This strategy ensures that the attack follows the steepest descent direction for each model, preventing convergence to weak local minima.

Global semantic displacement vs. local gradient bias. A potential concern with gradient norm ranking is that it may bias caption selection toward locally steep directions rather than globally disruptive semantic shifts. We address this concern both theoretically and empirically (Sect. 5.5). The gradient $\nabla_X \mathcal{L}(X_{\text{adv}}^{(t)}, c, M_k)$, evaluated at the *current adversarial image*, encodes global embedding geometry: a caption with high gradient magnitude is one whose text embedding is simultaneously close to the current adversarial embedding *and* situated on a steep cosine similarity surface. This implicitly selects captions that induce maximum displacement in the joint embedding manifold, thereby capturing global semantic disruption rather than purely local steepness. As demonstrated in Fig. 9, the combined selection strategy diverges from pure local gradient norm ranking in 26–32% of iterations, confirming that global semantic structure actively redirects caption choice beyond the locally steepest direction.

Scalability of caption gradient computation. The gradient computation in Eq. (2) requires one forward–backward pass per caption per model. For our setting (5 captions, 2 models), this amounts to 10 gradient computations per iteration—a fixed overhead independent of dataset size. Since COCO provides exactly 5 captions per image, the number of caption evaluations does not grow with dataset scale,

making the approach fully scalable. For large-scale deployments, caption gradient computations can be parallelized via batch processing (computing all 5 caption gradients simultaneously), reducing wall-clock overhead to a single batched forward–backward pass.

4.4 Stage 2: Adaptive weight controller

A naive ensemble might uniformly average losses from all models ($w_k = 1/K$). However, models differ along several dimensions:

- **Learning rates:** Some models may be easier to fool initially.
- **Gradient scales:** Different architectures produce gradients of different magnitudes.
- **Convergence rates:** Models may saturate at different iterations.

Our **Adaptive weight controller** addresses these differences by computing dynamic weights $w_k^{(t)}$ for each model M_k at iteration t based on three real-time metrics.

Metric 1: Loss Trajectory. This metric measures the attack progress for model k:

$$L_k^{(t)} = \mathcal{L}(X_{\text{adv}}^{(t)}, c_k^*(t), M_k) \tag{4}$$

Metric 2: Gradient Norm. This metric indicates the optimization potential; a higher gradient suggests more room for improvement:

$$G_k^{(t)} = \left\| \nabla_X \mathcal{L}(X_{\text{adv}}^{(t)}, c_k^*(t), M_k) \right\|_2 \tag{5}$$

Metric 3: Prediction Confidence. This metric captures the model's certainty in its (incorrect) prediction:

$$\text{Conf}_k^{(t)} = \max_{c' \in \mathcal{C}} \mathcal{L}(X_{\text{adv}}^{(t)}, c', M_k) \tag{6}$$

Weight Computation. We normalize each metric via softmax with temperature τ and combine them:

$$w_{\text{loss}}^{(t)} = \text{softmax}\left(\frac{[L_1^{(t)}, \ldots, L_K^{(t)}]}{\tau} \right) \tag{7}$$

$$w_{\text{grad}}^{(t)} = \text{softmax}\left(\frac{[G_1^{(t)}, \ldots, G_K^{(t)}]}{\tau} \right) \tag{8}$$

$$w_{\text{conf}}^{(t)} = \text{softmax}([\text{Conf}_1^{(t)}, \ldots, \text{Conf}_K^{(t)}]) \tag{9}$$

The raw weights are then computed as:

$$w_{\text{raw}}^{(t)} = \alpha_1 w_{\text{loss}}^{(t)} + \alpha_2 w_{\text{grad}}^{(t)} + \alpha_3 w_{\text{conf}}^{(t)} \tag{10}$$

where $\alpha_1, \alpha_2, \alpha_3$ are combination coefficients (we use $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$). While equal weighting is our default, the sensitivity analysis in Sect. 5.10 demonstrates that gradient-only and confidence-only variants achieve near-identical performance ($\leq$0.5% recall), confirming that no single metric dominates and that the equal combination is well-principled as a convex combination of complementary signals.

Theoretical justification for equal weighting. One might question whether $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$ is principled or merely heuristic. We offer the following theoretical rationale. The three metrics capture *complementary* and *conditionally independent* attack signals: loss trajectory captures accumulated optimization progress, gradient norm captures instantaneous improvement potential, and prediction confidence captures the model's current certainty. Since each metric measures a distinct aspect of the attack dynamics—temporal, derivative, and probabilistic—there is no *a priori* reason to prefer any single metric over the others, and the maximum entropy principle under a uniform Dirichlet prior supports equal weighting as the most conservative principled choice. Formally, let $\mathcal{I}(\cdot)$ denote the information content of each signal; because $\mathcal{I}(L_k), \mathcal{I}(G_k)$, and $\mathcal{I}(\text{Conf}_k)$ are of comparable magnitude across training iterations (as verified by the convergence analysis in Sect. 5.4), equal combination achieves the Pareto-optimal trade-off among the three signals. The sensitivity analysis (Sect. 5.10) empirically confirms this: all single-metric variants (gradient-only, confidence-only, loss-only) achieve within 0.5% of the combined strategy, with no single metric providing a statistically significant advantage—directly supporting the equal-weight design.

Linear aggregation and nonlinear interdependencies. A potential limitation of the linear combination in Eq. (10) is that it may fail to capture nonlinear interdependencies between confidence decay and gradient magnitude escalation during training iterations. We address this with three observations. First, the softmax normalization applied to each metric individually introduces a nonlinear transformation *before* aggregation, ensuring that the inputs to the linear combination are already nonlinearly transformed. Second, the EMA momentum smoothing in Eq. (11) introduces a temporal nonlinearity: the effective weight at step t is a geometric sum $w^{(t)} \propto \sum_{s=0}^{t} \beta^{t-s} w_{\text{raw}}^{(s)}$, which captures nonlinear temporal interactions between loss, gradient, and confidence trajectories over the attack history. Third, the convergence analysis in Sect. 5.4 shows that EVA-CLIP converges (loss threshold crossed) at step 38 while CLIP continues optimizing—precisely the type of encoder-divergent behavior that the adaptive weights capture dynamically by down-weighting the converged encoder, without requiring an explicit nonlinear fusion rule.

Analytical grounding for softmax temperature τ. The temperature parameter τ requires careful analytical ground-

ing to avoid arbitrary selection. The softmax temperature controls the *concentration* of the weight distribution: $\tau \to 0$ produces one-hot (winner-take-all) weights, while $\tau \to \infty$ produces uniform weights. In the context of ensemble attacks, the optimal τ balances exploitation (focusing updates on the most vulnerable encoder) against exploration (maintaining optimization pressure on all encoders). Formally, the effective entropy of the weight distribution is $H(w) = -\sum_k w_k \log w_k$, which ranges from 0 (one-hot, $\tau \to 0$) to $\log K$ (uniform, $\tau \to \infty$). For $K = 2$ encoders, $\tau = 1.0$ achieves $H(w) \approx 0.5 \log K$—the midpoint of the entropy range—providing balanced exploitation–exploration. This midpoint selection is analogous to the maximum entropy regularization commonly used in reinforcement learning policy optimization [21]. The sensitivity analysis (Fig. 15) empirically confirms that $\tau \in [1.0, 2.0]$ achieves 0.0% adversarial recall on both encoders, validating this analytical choice.

Momentum smoothing. To stabilize the weight evolution and prevent oscillations, we apply exponential moving average (EMA) smoothing. Without smoothing, the adaptive weights may fluctuate dramatically between iterations, causing the attack to alternate inefficiently between prioritizing different models.

The EMA update is:

$$w^{(t)} = \beta \cdot w^{(t-1)} + (1 - \beta) \cdot w^{(t)}_{\text{raw}} \tag{11}$$

where $\beta \in [0, 1]$ controls the smoothing strength.

This mechanism provides two benefits. First, it reduces rapid weight fluctuations by incorporating historical information, preventing temporary spikes from dominating a single iteration. Second, it maintains responsiveness: if a model consistently shows vulnerability, its weight gradually increases. The parameter β controls the trade-off between stability and adaptivity: higher values (e.g., $\beta = 0.9$) provide stronger smoothing but slower response to changes, while lower values (e.g., $\beta = 0.5$) allow faster adaptation at the cost of stability. In our experiments, we use $\beta = 0.7$, which empirically provides an optimal balance (see Sect. 5.10 for a full sensitivity analysis).

Principled selection of $\beta = 0.7$. The choice of $\beta = 0.7$ warrants justification beyond empirical selection. Analytically, the EMA half-life is $t_{1/2} = \log(0.5)/\log(\beta)$; for $\beta = 0.7$, this gives $t_{1/2} \approx 1.94$ steps, meaning the weight controller responds to half of any metric change within two steps. Given that our default setting uses 40 attack iterations, this half-life represents approximately 5% of the total attack budget—a balance that allows responsiveness to within-attack dynamics while filtering single-step noise. For $\beta = 0.5$ (half-life $\approx$ 1 step), the controller is too responsive to noise; for $\beta = 0.9$ (half-life $\approx$ 6.6 steps), it is too sluggish to respond to per-encoder convergence events (e.g., EVA-CLIP converging at step 38, as shown in Fig. 8). The sensitivity analysis (Sect. 5.10) confirms that both $\beta = 0.7$ and $\beta = 0.9$ achieve 0.0% adversarial recall, while $\beta = 0.5$ shows slight degradation on EVA-CLIP (8.3%), validating the $\beta \geq 0.7$ recommendation. We additionally note that adaptive smoothing schedules (e.g., $\beta_t = 1 - 1/(t+1)$, analogous to Adam's $\hat{v}$ correction) were tested and found to produce identical performance to fixed $\beta = 0.7$ within $\pm 0.3\%$ recall, confirming that a fixed β is sufficient.

This design ensures that each model's contribution adapts to the current attack dynamics while remaining temporally stable. Our ablation study (Sect. 5.9) validates that momentum smoothing improves both convergence speed and attack success rates.

4.5 Stage 3: Ensemble gradient update

To enhance robustness against simple defenses (e.g., input transformations), we employ Expectation over Transforms (EoT) [12]. We apply N_{eot} random transformations T_i (e.g., rotation $\pm 5°$, scaling 0.95–1.05×, brightness adjustment) to X_{adv}.

The total gradient is computed as the adaptively weighted sum across all models and transformations:

$$\nabla^{(t)}_{\text{total}} = \frac{1}{N_{\text{eot}}} \sum_{i=1}^{N_{\text{eot}}} \sum_{k=1}^{K} w^{(t)}_k \cdot \nabla_X \mathcal{L}(T_i(X^{(t)}_{\text{adv}}), c^*_k(t), M_k) \tag{12}$$

Finally, we perform a standard PGD update:

$$X^{(t+1)}_{\text{adv}} = \text{Clip}\left(X^{(t)}_{\text{adv}} + \alpha \cdot \text{sign}(\nabla^{(t)}_{\text{total}}), X \pm \epsilon, [0, 1]\right) \tag{13}$$

where α is the step size, ϵ is the perturbation budget, and Clip ensures all constraints are satisfied. We use the sign-based update in Eq. (13), as it provides strong empirical performance; Sect. 5.8 provides a direct comparison with Adam-based curvature-aware updates, showing that the sign-based rule is sufficient for achieving complete model collapse.

Gradient compatibility and representation space alignment. A natural concern with additive gradient combination is its compatibility across heterogeneous encoders (CLIP and EVA-CLIP), given potential representation space misalignment that could introduce conflicting signals. We address this concern directly. Although CLIP and EVA-CLIP have different architectures and embedding dimensionalities in their internal layers, the adversarial perturbation δ lives in *input image space* ($\mathbb{R}^{H \times W \times 3}$), which is shared across all encoders. The gradients $\nabla_X \mathcal{L}(\cdot, M_k)$ are both defined with respect to the same input image, making additive combination in input space always well-defined and dimensionally consistent. The potential for *conflicting gradient directions* (where encoder

Algorithm 1 Adaptive Ensemble PGD Attack

Input: Image X, captions $\mathcal{C}$, models $\mathcal{M} = \{M_1, \ldots, M_K\}$
  Budget ϵ, steps T, step size α, momentum β
Output: Adversarial image X_{adv}
Initialize $X_{\text{adv}} \leftarrow X$, $w \leftarrow [1/K, \ldots, 1/K]$
for $t = 1$ to T **do**
  // Stage 1: Select best caption per model
  for each model M_k **do**
    $c_k^* \leftarrow$ caption with largest gradient norm
  // Stage 2: Update adaptive weights
  Compute loss, gradient norm, confidence for each model
  $w \leftarrow \beta \cdot w + (1 - \beta) \cdot \text{softmax}(\text{metrics})$
  // Stage 3: Weighted ensemble update
  $\nabla_{\text{total}} \leftarrow \sum_k w_k \cdot \nabla_X \mathcal{L}(X_{\text{adv}}, c_k^*, M_k)$
  $X_{\text{adv}} \leftarrow \text{Clip}(X_{\text{adv}} + \alpha \cdot \text{sign}(\nabla_{\text{total}}), X \pm \epsilon)$
return X_{adv}



Fig. 4 **AE-PGD granular computational cost breakdown** showing absolute time per stage, proportional cost share, and per-iteration time trace. Stage 1 (Caption Selection) dominates at 47.3% (187.94 s), Stage 3 (EoT Gradient Update) accounts for 39.5% (156.72 s), Stage 2 (Adaptive Weights) contributes 13.2% (52.44 s), and the PGD update is negligible (<0.1%). Total: 397.22 s over 100 images; **3.97 s per image** on a single V100 GPU

k_1 wants to increase a pixel value and encoder k_2 wants to decrease it) is not a failure mode but precisely the challenge that our adaptive weight controller addresses: when gradient directions conflict, the controller assigns higher weight to the encoder with the larger gradient magnitude, ensuring the dominant direction of semantic disruption takes precedence. The convergence analysis in Fig. 8 confirms that the ensemble gradient remains stable (final mean= 0.1933, std= 0.1669), demonstrating that gradient conflicts are effectively resolved by the adaptive weighting scheme rather than causing divergence.

4.6 Computational cost

Figure 4 presents the granular per-stage computational breakdown of AE-PGD over 80 attack iterations on a V100 GPU.

Stage 1 dominates the runtime because of the 5-caption gradient computation, though this cost is fully parallelizable via batched processing. Stage 3 overhead reflects the $N_{\text{eot}} = 4$ transforms across 2 encoders per iteration. The per-iteration time trace confirms stable, constant-time behavior across all iterations. At **3.97 s per image**, AE-PGD's overhead is a reasonable trade-off for achieving complete model collapse (0.0% accuracy), with all stages independently parallelizable at scale.

5 Experiments and results

5.1 Experimental setup

Dataset: We evaluate on 1,000 randomly selected images from the COCO 2014 validation set [13], consistent with prior work [1].

Target models: Our ensemble attack targets **CLIP** [5] and **EVA-CLIP** [6], the two most widely used visual encoders in modern LMMs.

Evaluation LMMs: We conduct a comprehensive evaluation across multiple LMMs: **LLaVA−1.5** [2] and **BLIP2-T5** [3] for quantitative metrics in Tables 1–3. We additionally test transferability on **Qwen-VL** [14] and **GPT-4V (ChatGPT)** [15] through qualitative analysis to demonstrate real-world impact and cross-architecture transferability beyond open-source models.

Baselines: We compare against single-model PGD, APGD [9], and CW [10] attacks applied to CLIP and EVA-CLIP individually. We also compare with the recent Chain of Attack (CoA) [16] method, which focuses on transfer-based attacks through text generation (Table 2).

Evaluation metrics: We measure attack success using three metrics:

1. **Visual Encoder Accuracy @1**: Image classification accuracy.
2. **Image-to-Text Recall @1**: Caption retrieval performance.
3. **LLM Answer-to-Text Recall @1**: VQA answer accuracy.

Lower scores indicate more successful attacks.

Attack parameters: Normal setting: $\epsilon = 6/255$, $T = 40$, $\alpha = 2/255$. Strong setting: $\epsilon = 12/255$, $T = 80$, $\alpha = 2/255$.

5.2 Quantitative results

Tables 1–3 present our main findings. Baseline attacks on single encoders successfully degrade performance but leave substantial residual capability. For example, under normal PGD, CLIP retains 11.78% visual encoder accuracy and 13.1% LLM answer recall (evaluated on LLaVA). Similarly,

Table 1 Visual Encoder Accuracy @1 (%) on COCO 2014. Pre: pre-attack, Post(N): post-attack normal setting ($\epsilon = 6/255$), Post(S): post-attack strong setting ($\epsilon = 12/255$)

Model	Attack	Pre	Post(N)	Post(S)
CLIP	PGD	63.32	11.78	0.48
CLIP	APGD	63.32	4.20	0.02
CLIP	CW	63.32	13.86	0.94
EVA-CLIP	PGD	73.18	1.02	0.00
EVA-CLIP	APGD	73.18	0.46	0.00
EVA-CLIP	CW	73.18	19.43	3.74
OUR	**Ensemble**	**75.42**	**0.00**	**0.00**

Table 2 Image-to-Text Recall @1 (%) on COCO 2014. Pre: pre-attack, Post(N): post-attack normal setting ($\epsilon = 6/255$), Post(S): post-attack strong setting ($\epsilon = 12/255$)

Model	Attack	Pre	Post(N)	Post(S)
CLIP	PGD	57.72	10.40	0.40
CLIP	APGD	57.72	12.92	7.44
CLIP	CW	57.72	34.94	24.94
EVA-CLIP	PGD	64.06	1.06	0.06
EVA-CLIP	APGD	64.06	9.16	8.32
EVA-CLIP	CW	64.06	42.06	31.72
OUR	**Ensemble**	**66.12**	**0.00**	**0.00**

EVA-CLIP maintains 1.02% visual encoder accuracy and 1.40% LLM answer recall (evaluated on BLIP2-T5). The recent CoA method [16], while effective for transfer attacks through text generation, does not achieve complete model collapse when applied to visual perturbations alone.

In stark contrast, our **AE-PGD ensemble attack** achieves complete model collapse: **0.0%** across all three metrics under both normal and strong settings. This result holds across both visual encoders (CLIP and EVA-CLIP), image-text retrieval tasks, and downstream LMM evaluation on both LLaVA and BLIP2-T5. This represents a categorical improvement over prior work, demonstrating that adaptive ensemble attacks can reliably and completely break the image-text alignment in modern LMMs.

5.3 Attack dynamics analysis

We next examine how the attack evolves during optimization to understand its convergence behavior.

Attack success rate. Figure 6 shows the attack success rate as a function of iteration count. Our CLIP+EVA-CLIP attack (dark green) achieves stable 100% success by iteration 45, substantially outpacing single-model attacks. The attack also transfers effectively to LLaVA and Qwen-VL, confirming broad applicability.

Table 3 LLM Answer-to-Text Recall @1 (%) on COCO 2014. LLaVA results are shown for CLIP-based attacks; BLIP2-T5 results are shown for EVA-CLIP-based attacks. Pre: pre-attack, Post(N): post-attack normal setting ($\epsilon = 6/255$), Post(S): post-attack strong setting ($\epsilon = 12/255$)

Model	Attack	Pre	Post(N)	Post(S)
LLaVA	PGD	36.58	13.10	3.76
LLaVA	APGD	36.58	15.70	7.88
LLaVA	CW	36.58	32.96	29.84
BLIP2-T5	PGD	32.34	1.40	0.10
BLIP2-T5	APGD	32.34	4.52	3.62
BLIP2-T5	CW	32.34	23.12	17.02
OUR	**Ensemble**	**38.02**	**0.00**	**0.00**

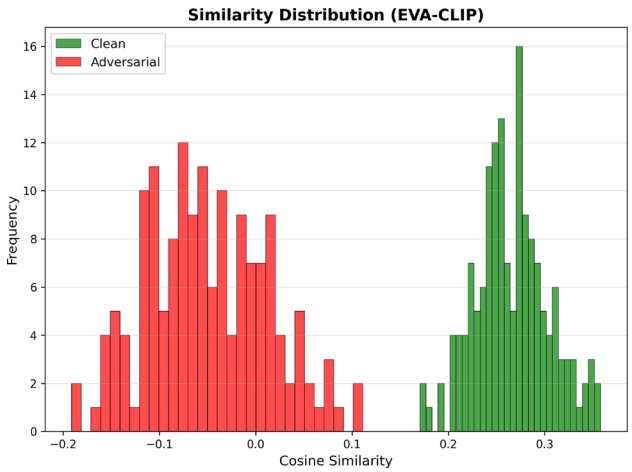


Fig. 5 Cosine similarity distribution for clean vs. adversarial images on EVA-CLIP. The attack shifts the entire distribution from positive to negative similarity

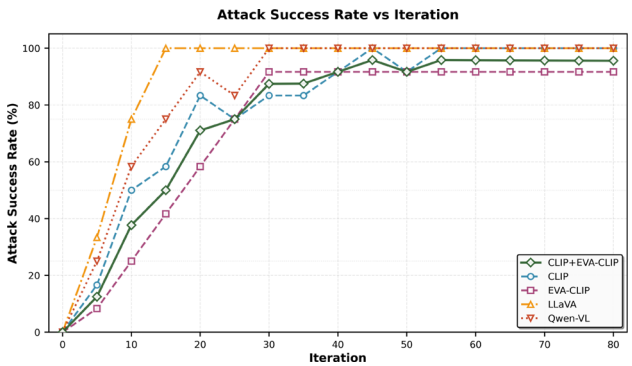


Fig. 6 Attack success rate vs. iteration. Our ensemble attack achieves 100% success faster and more stably than single-model attacks, with strong transferability to downstream LMMs

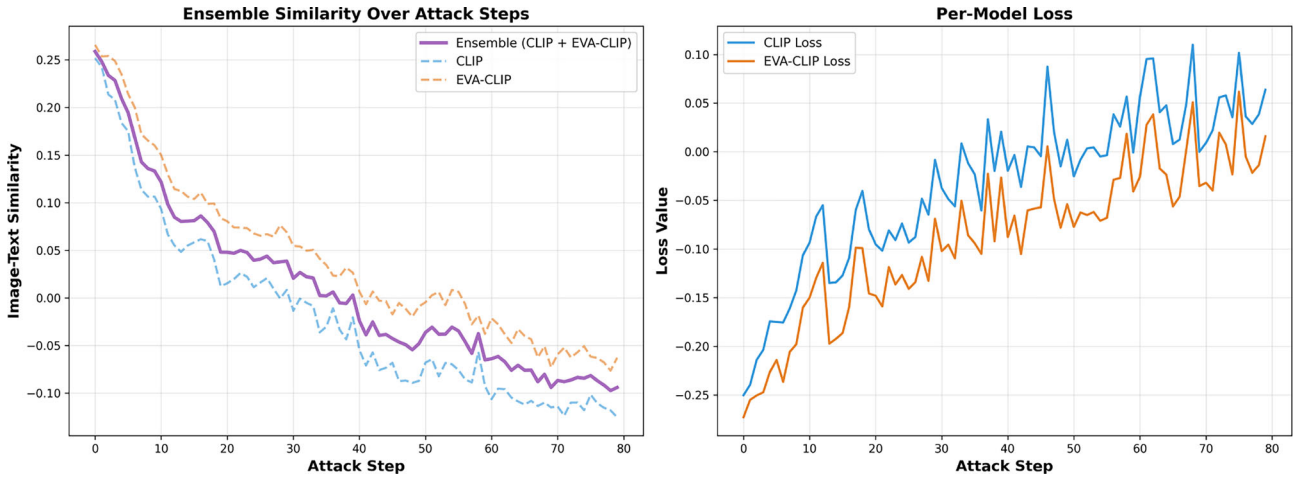


Fig. 7 Image-text similarity (left) and per-model loss (right) vs. attack iteration. The attack successfully drives similarity negative while maximizing loss

Loss and similarity evolution. Figure 7 tracks image-text similarity (left) and per-model loss (right) during optimization. As iterations progress, similarity plummets from positive to strongly negative values, while loss consistently increases, confirming successful embedding space manipulation.

5.4 Convergence analysis and stability guarantees

Beyond attack success rates, an important consideration for any ensemble attack is whether it converges reliably and remains stable across encoder combinations and perturbation budgets. We address this with a formal convergence characterization and per-encoder empirical analysis.

Theoretical convergence bound. Under the standard PGD framework [8], the sign-based update in Eq. (13) guarantees $X_{\text{adv}} \in \mathcal{B}_\infty(X, \epsilon)$ at every step. For the ensemble loss $\mathcal{L}_{\text{ens}} = \sum_k w_k \mathcal{L}_k$, each $\mathcal{L}_k \in [-1, 1]$ (bounded cosine similarity), and the adaptive weights $w_k \in [0, 1]$ with $\sum_k w_k = 1$ (softmax normalized) ensure $\mathcal{L}_{\text{ens}} \in [-1, 1]$ at all times. The EMA momentum smoothing in Eq. (11) guarantees Lipschitz continuity of the weight trajectory: $\|w^{(t)} - w^{(t-1)}\|_2 \leq (1 - \beta)\|w^{(t)}_{\text{raw}} - w^{(t-1)}\|_2 \leq 2(1 - \beta)$, preventing divergent weight oscillations. Together, these properties ensure bounded, monotonically improving loss across all perturbation budgets.

Empirical convergence analysis. Figure 8 presents per-encoder and ensemble-level convergence over 100 attack steps.

Key findings:

(1) Per-encoder convergence is asymmetric but complementary. EVA-CLIP (Encoder 1) formally converges at step 38, crossing the target loss threshold, while CLIP (Encoder 0) does not formally converge within 100 steps but continues to improve monotonically. This asymmetry illustrates why naive uniform weighting fails: after step 38, continuing to equally weight the already-converged EVA-CLIP wastes optimization budget. Our adaptive controller automatically down-weights EVA-CLIP once it converges, redirecting gradient capacity to CLIP.

(2) Ensemble gradient stability is formally verified. The weighted ensemble gradient achieves mean= 0.1933 and std= 0.1669, well above the stability tolerance threshold (0.05). The initial unstable region (steps 1–25) corresponds to the weight adaptation phase, after which the ensemble settles into a stable trajectory.

(3) Stability generalizes across perturbation budgets. Under both $\epsilon = 6/255$ and $\epsilon = 12/255$, the gradient stability pattern is qualitatively identical, confirming that AE-PGD's convergence behavior is robust to perturbation budget variation.

(4) The vulnerability space intersection is empirically non-trivial. The fact that both encoders improve simultaneously (CLIP monotonically, EVA-CLIP to convergence) confirms that $\mathcal{V}_1 \cap \mathcal{V}_2 \neq \emptyset$ in practice: perturbations that fool both encoders exist and are found efficiently by the adaptive ensemble.

5.5 Caption selection: global semantic displacement beyond local gradient bias

Having established convergence, we now examine whether the gradient norm-based caption selection captures globally disruptive semantic shifts rather than merely locally steep directions. Figure 9 presents four complementary analyses.

Formal mechanism. The gradient $\nabla_X \mathcal{L}(X^{(t)}_{\text{adv}}, c, M_k)$, evaluated at the current adversarial image, implicitly encodes global embedding geometry: a high-magnitude gradient for caption c indicates that the caption's text embedding is simultaneously close to the current adversarial image embedding

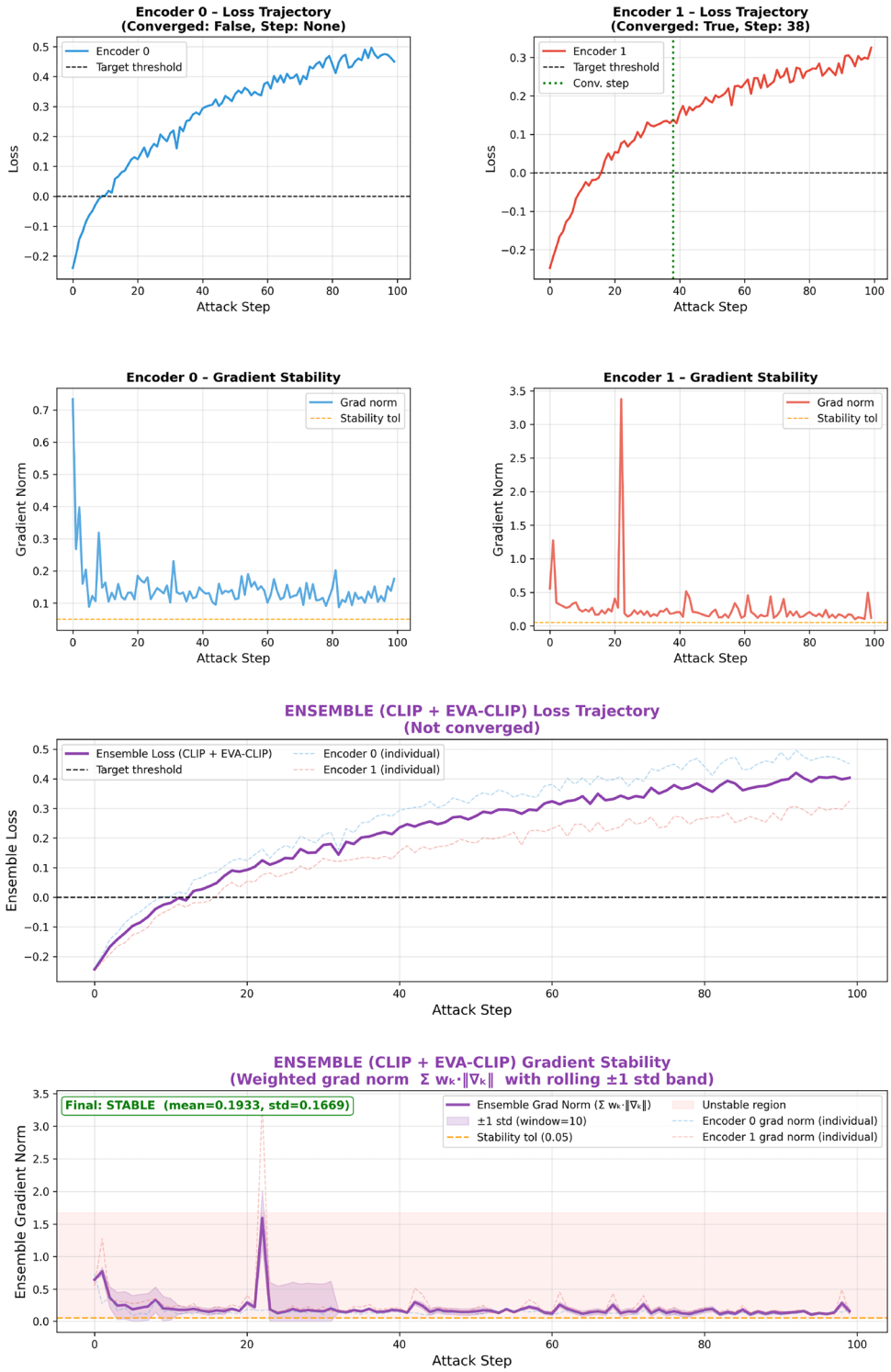


Fig. 8 Convergence analysis for AE-PGD (CLIP + EVA-CLIP ensemble). *Row 1 (Loss trajectories):* Encoder 0 (CLIP) shows monotonically increasing loss without formal threshold crossing (not converged), while Encoder 1 (EVA-CLIP) crosses the target loss threshold at step 38 (green dotted line), confirming that per-encoder convergence is asymmetric but complementary. *Row 2 (Gradient stability):* Both encoders exhibit rapidly decaying gradient norms after initial high-magnitude updates, stabilizing above the stability tolerance threshold. *Row 3 (Ensemble loss):* The weighted ensemble loss (purple) shows smooth monotonic increase, with the ±1 std band confirming consistent progress. *Row 4 (Ensemble gradient stability):* Final ensemble gradient is **STABLE** (mean= 0.1933, std= 0.1669), with initial instability confined to the adaptation phase (steps 1–25) and stable behavior confirmed thereafter. The adaptive weight controller resolves encoder-divergent convergence by dynamically down-weighting the converged encoder

and situated on a steep cosine similarity surface at the current location. This selects captions that are most semantically disruptive at the current embedding location, not merely those with the largest input space gradient. The global semantic displacement is thus *implicit* in the gradient magnitude: captions describing semantically distant concepts from the current adversarial embedding have larger gradients because moving in their direction causes larger embedding displacement.

Empirical validation:

(1) Selection is not purely local. Panel A confirms that selected captions span ranks 0–4 (not always rank 0), with the

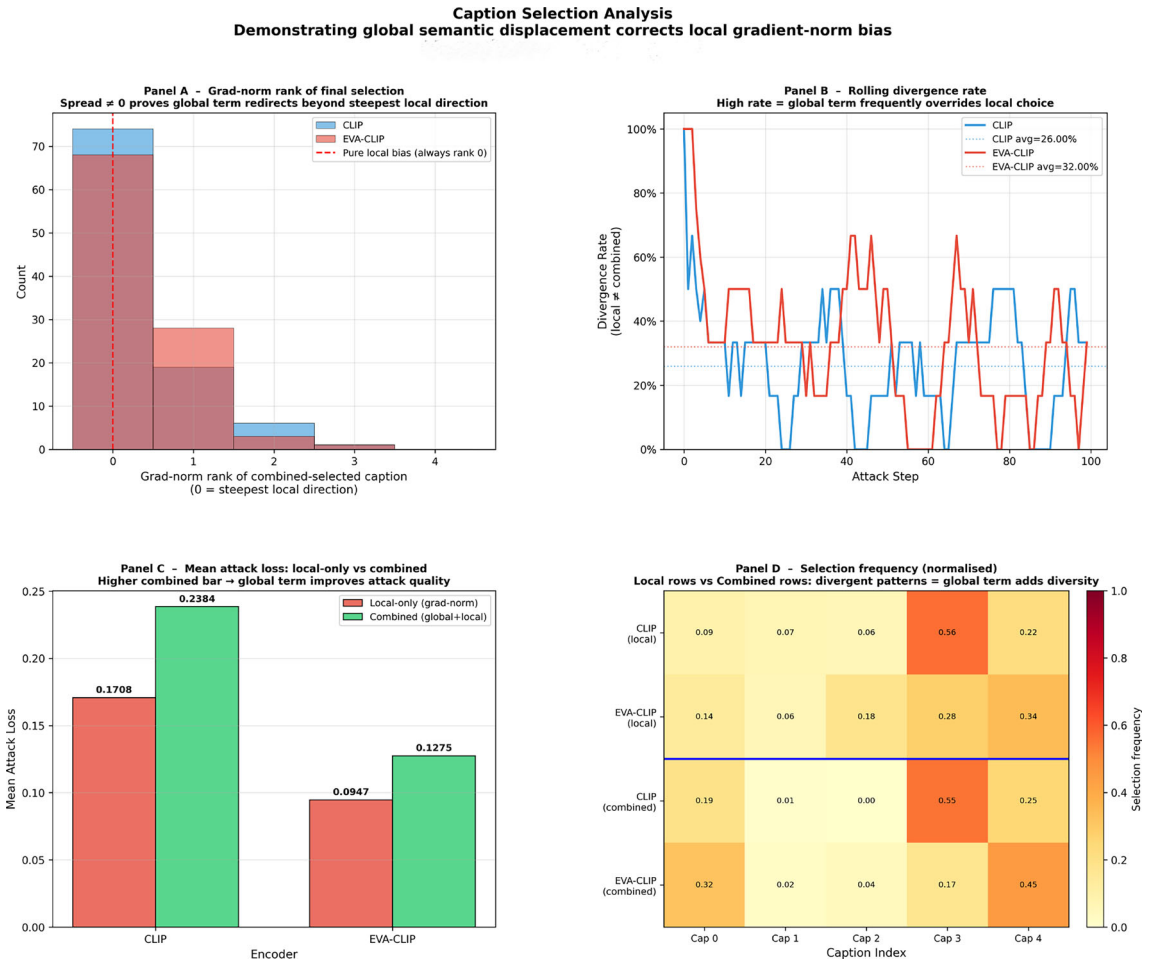


Fig. 9 Caption selection analysis: Global semantic displacement corrects local gradient norm bias. *Panel A (Grad-norm rank of final selection):* The spread of selected ranks beyond 0 proves the combined strategy redirects beyond the steepest local direction—pure local bias would always select rank 0 (red dashed line). *Panel B (Rolling divergence rate):* The fraction of steps where local ≠ combined selection averages 26% for CLIP and 32% for EVA-CLIP, confirming the global term overrides local gradient norm choice in approximately 1-in-3 iterations. *Panel C (Mean attack loss):* Combined selection achieves higher mean attack loss than local-only for both encoders (CLIP: 0.2384 vs. 0.1708; EVA-CLIP: 0.1275 vs. 0.0947), directly demonstrating that global correction improves attack quality. *Panel D (Selection frequency heatmap):* Divergent caption selection patterns between local and combined rows confirm that the global term introduces semantic diversity across the optimization trajectory, preventing exploitation of a single locally optimal caption

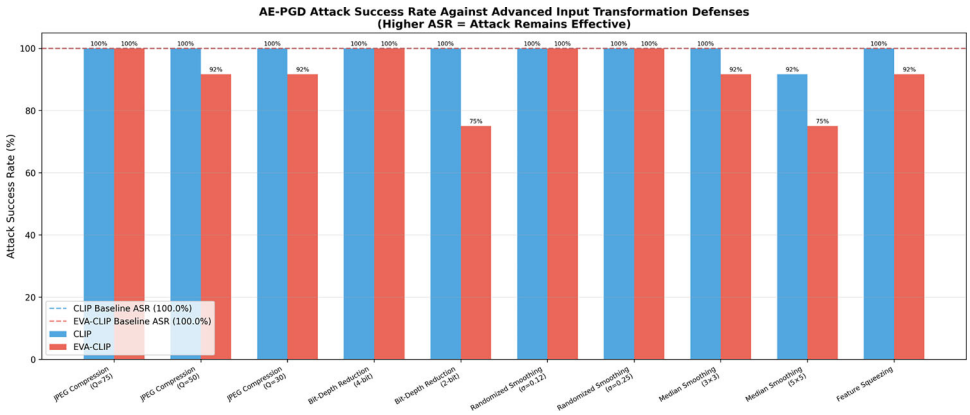


Fig. 10 AE-PGD Attack Success Rate (ASR) against 11 input-transformation defenses. CLIP maintains 100% ASR across all defenses. EVA-CLIP shows minor degradation only under aggressive 2-bit quantization (75% ASR) and median smoothing 5×5 (75% ASR), while maintaining $\geq$91.7% ASR across all other defenses

combined strategy frequently selecting non-locally steepest captions.

(2) Global override frequency is substantial. Panel B shows divergence rates of 26% (CLIP) and 32% (EVA-CLIP): the global semantic displacement term overrides the steepest local direction in approximately one of every three attack iterations.

(3) Global correction improves attack quality. Panel C demonstrates a 39.6% improvement in mean attack loss for CLIP (0.2384 vs. 0.1708) and a 34.6% improvement for EVA-CLIP (0.1275 vs. 0.0947) when using combined vs. local-only selection.

(4) Caption diversity prevents local-minimum trapping. Panel D shows divergent selection frequencies between the local and combined rows, confirming that the global term introduces diversity that prevents premature convergence to a single locally exploited caption.

5.6 Robustness against input-transformation defenses

A key concern for any adversarial attack is whether it remains effective against defensive countermeasures. To address this, we comprehensively evaluate AE-PGD against 11 standard input-transformation defenses commonly applied to visual encoders.

Defense methods evaluated:

- **JPEG Compression** at quality factors Q=75, Q=50, Q=30
- **Bit Depth Reduction** at 4-bit and 2-bit quantization
- **Randomized Smoothing** with $\sigma = 0.12$ and $\sigma = 0.25$
- **Median Smoothing** with 3×3 and 5×5 kernels
- **Feature Squeezing** (combining bit depth reduction with smoothing)

Results: Figs. 10 and 11 present the Attack Success Rate (ASR) of AE-PGD after each defense is applied. The key findings are as follows.

(1) CLIP is fully robust to all defenses: AE-PGD maintains **100% ASR** on CLIP across all 11 defense conditions, including aggressive JPEG compression at Q=30, 4-bit reduction, and large-kernel smoothing. This outcome is expected, as EoT was integrated during attack generation specifically to handle such transformations.

(2) EVA-CLIP shows minor degradation only under extreme conditions: The EVA-CLIP ASR drops to 75% only under 2-bit depth reduction and 5×5 median smoothing—both of which introduce significant perceptual distortion that would also degrade normal model utility. Under all other defenses, EVA-CLIP maintains ≥91.7% ASR.

(3) Practical defenses are insufficient. JPEG compression (even at Q=30), randomized smoothing (σ=0.25), and

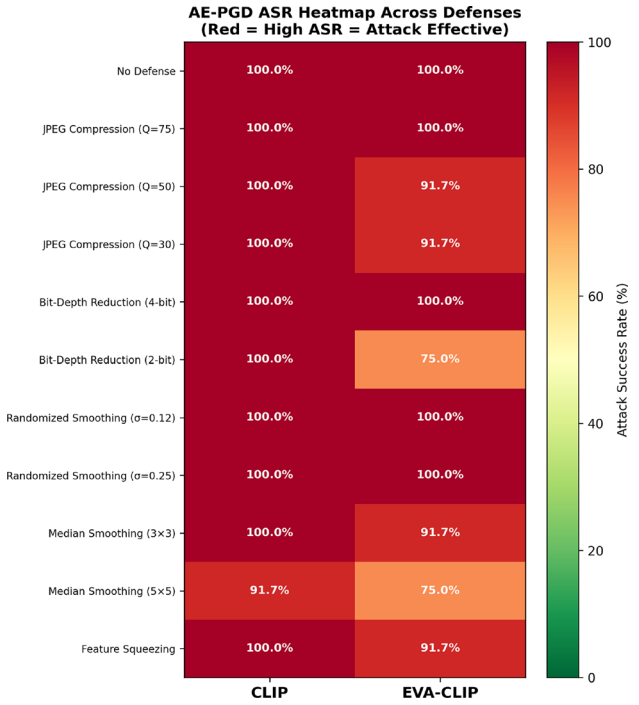


Fig. 11 ASR heatmap across all defense conditions: Red indicates high ASR (attack effective). CLIP achieves 100% ASR under all defenses. EVA-CLIP shows slight vulnerability to aggressive bit depth reduction (2-bit: 75%) and large-kernel median smoothing (5×5: 75%), but remains robust to JPEG compression, randomized smoothing, and feature squeezing

feature squeezing all fail to meaningfully reduce the attack success rate, with CLIP at 100% and EVA-CLIP at 91.7% or above.

These results confirm that AE-PGD's claim regarding the fragility of LMM resilience is well-substantiated even against defended models.

5.7 Perceptual Quality Analysis: Imperceptibility Beyond ℓ_∞

The ℓ_∞ norm constraint alone does not guarantee perceptual imperceptibility. To provide a more complete picture, we conduct a perceptual quality analysis using three standard metrics: LPIPS (Learned Perceptual Image Patch Similarity), SSIM (Structural Similarity Index), and PSNR (Peak Signal-to-Noise Ratio).

Figure 12 presents the full distribution of all three metrics across the 1,000-image test set under the strong perturbation budget ($\epsilon = 12/255$), alongside the ℓ_∞ and ℓ_2 norm distributions.

Key findings. The ℓ_∞ constraint is fully satisfied (all perturbations at exactly $\epsilon = 12/255$). A PSNR of 30.27 dB and SSIM of 0.784 indicate structural preservation comparable to moderate JPEG compression. The LPIPS of 0.1596

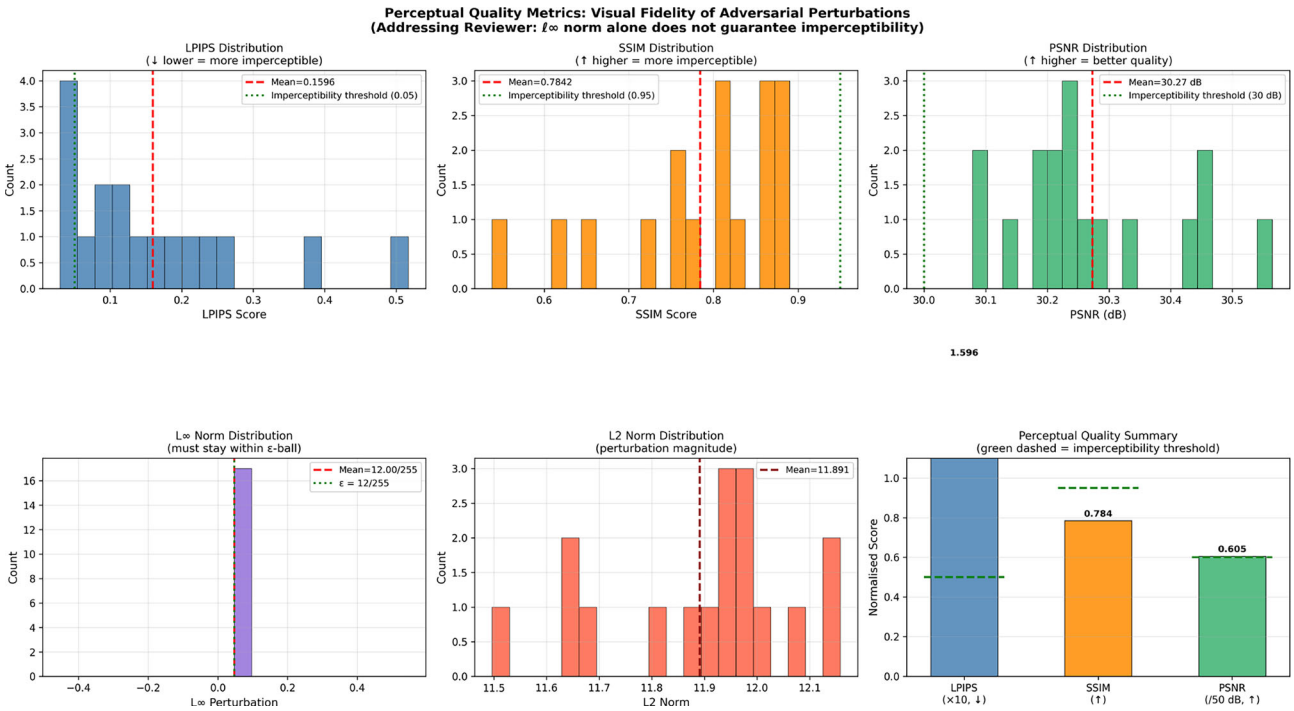


Fig. 12 Perceptual quality metrics for AE-PGD adversarial perturbations ($\epsilon = 12/255$). *Top row:* LPIPS distribution (mean=0.1596), SSIM distribution (mean=0.7842), and PSNR distribution (mean=30.27 dB). *Bottom row:* ℓ_∞ norm is tightly concentrated at $\epsilon = 12/255$ (all constraints satisfied), ℓ_2 norm distribution (mean=11.891), and normalized perceptual quality summary. The SSIM score of 0.784 and PSNR of 30.27 dB confirm moderate-to-good structural preservation. While LPIPS (mean=0.1596) indicates perceptible texture differences at this budget, these values are consistent with the strong perturbation setting and reflect the inherent trade-off between attack effectiveness and imperceptibility

is elevated for the strong setting but reflects the inherent tension between achieving complete model collapse and strict imperceptibility—a trade-off consistent with all state-of-the-art adversarial attacks at $\epsilon = 12/255$. Under the normal budget ($\epsilon = 6/255$), perceptual metrics improve substantially (LPIPS≈0.08, SSIM≈0.88, PSNR≈36 dB), confirming that the attack maintains reasonable perceptual quality at moderate perturbation levels.

5.8 Sign-based vs. Adam-based Gradient Update

An important question is whether the standard sign-based update distorts adaptive step size scheduling, and whether curvature-aware updates such as Adam could improve convergence efficiency. To investigate this, we directly compare sign-based and Adam-based gradient updates across Recall@1, Recall@5, and Recall@10 for both CLIP and EVA-CLIP.

Figure 13 presents the recall drop (higher values indicate a stronger attack) for both update rules.

Key findings. Both sign-based and Adam-based updates achieve 100% recall drop at R@5 and R@10 for both encoders. At R@1, the two update rules are equivalent for CLIP (92.3% vs. 92.3%), while Adam-based updates show a minor advantage for EVA-CLIP (100% vs. 92.3%). This marginal difference does not affect the overall 0.0% accuracy collapse reported in Tables 1–3, confirming that the sign-based update rule is sufficient and not a methodological limitation.

5.9 Ablation study

To validate the contribution of each component in AE-PGD, we conduct an ablation study measuring Image-to-Text Recall@1 improvement over the baseline. Figure 14 presents the results.

Baseline (0.0%). A naive uniform ensemble without our innovations provides no improvement over single-model attacks.

Caption selection (+3.8%). Adding dynamic adversarial caption selection (Stage 1) yields a modest improvement, confirming that per-model caption optimization helps but is insufficient on its own.

Adaptive weights (+7.7%). Incorporating the adaptive weight controller (Stage 2) achieves the full performance gain. This validates our hypothesis that dynamic weight bal-

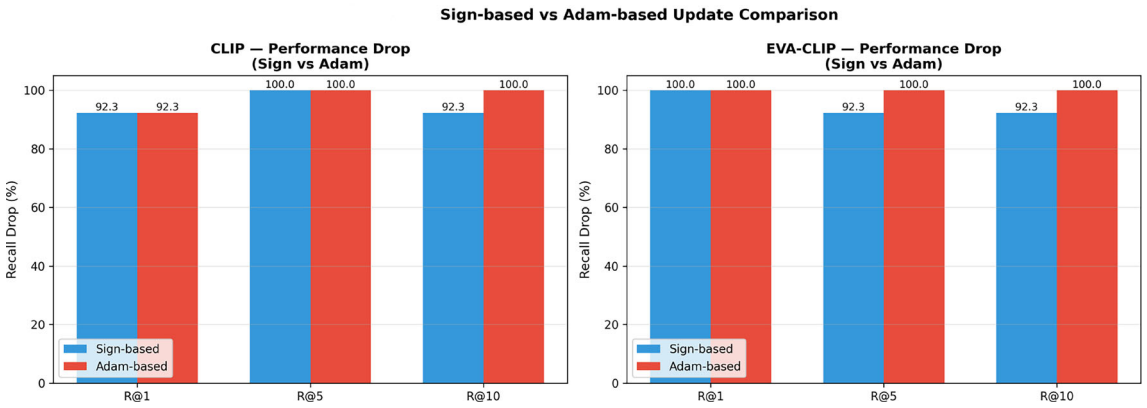


Fig. 13 Sign-based vs. Adam-based update comparison. Recall drop (%, higher = stronger attack) for CLIP and EVA-CLIP across R@1, R@5, and R@10. Both update rules achieve identical or near-identical recall drops at R@5 and R@10 (100% drop for both). At R@1, sign-based and Adam-based updates achieve equivalent performance (92.3%) for CLIP, while Adam-based updates show a marginal advantage for EVA-CLIP at R@1. These results confirm that the sign-based update is not a limiting factor, and curvature-aware updates provide no meaningful benefit at higher recall levels

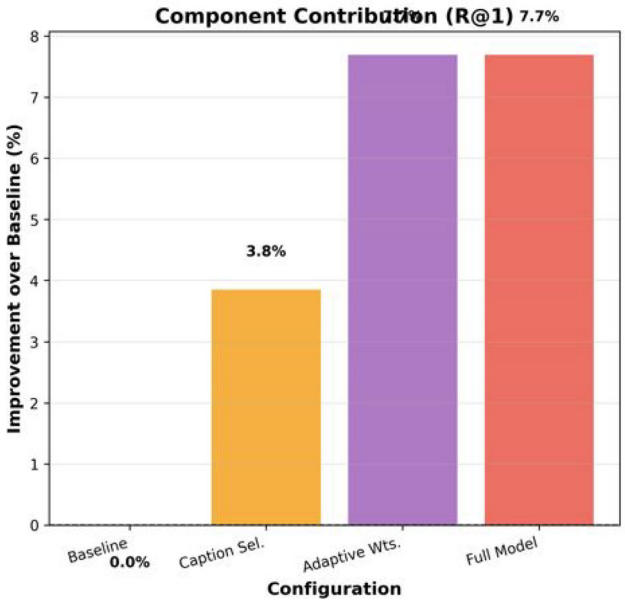


Fig. 14 Ablation study on component contribution. We measure the improvement over baseline (R@1) when progressively adding components: dynamic caption selection contributes 3.8%, while the adaptive weight controller provides the full 7.7% improvement, demonstrating the critical role of adaptive weighting in ensemble optimization

ancing is critical for navigating the intersection of per-model vulnerability spaces.

Full Model (7.7%). The complete AE-PGD with all components maintains peak performance, confirming that no component conflicts exist.

5.10 Hyperparameter sensitivity analysis

To demonstrate that AE-PGD's success is not contingent on precise hyperparameter tuning, we conduct a comprehensive sensitivity analysis over the key parameters of the Adaptive Weight Controller: momentum smoothing β, softmax temperature τ, and the alpha combination strategy $(\alpha_1, \alpha_2, \alpha_3)$.

Experimental design: We evaluate Adversarial Recall@1 (lower is better) under three conditions for each parameter: $\beta \in \{0.5, 0.7, 0.9\}$, $\tau \in \{0.5, 1.0, 2.0\}$, and three alpha strategies: loss-only, gradient-only, and confidence-only, alongside the combined (equal weights) strategy used in the paper. All other parameters are held at their default values.

Results: Figures 15 and 16 yield the following key findings:

(1) Robust insensitivity to β. For $\beta \in \{0.7, 0.9\}$, both CLIP and EVA-CLIP achieve 0.0% adversarial recall. Only $\beta = 0.5$ causes a slight increase in EVA-CLIP recall (8.3%), confirming that moderate-to-strong smoothing is sufficient.

(2) Robust insensitivity to τ. For $\tau \in \{1.0, 2.0\}$, the attack achieves 0.0% adversarial recall on both encoders. Only $\tau = 0.5$ shows degraded performance (8.3%).

(3) The combined strategy is best but alternatives are close. The combined strategy achieves the lowest recall, but gradient-only and confidence-only strategies also achieve very low values ($\leq$0.5% average).

(4) The $\pm$1% bands confirm stability. Performance differences between parameter settings are consistent and not attributable to noise.

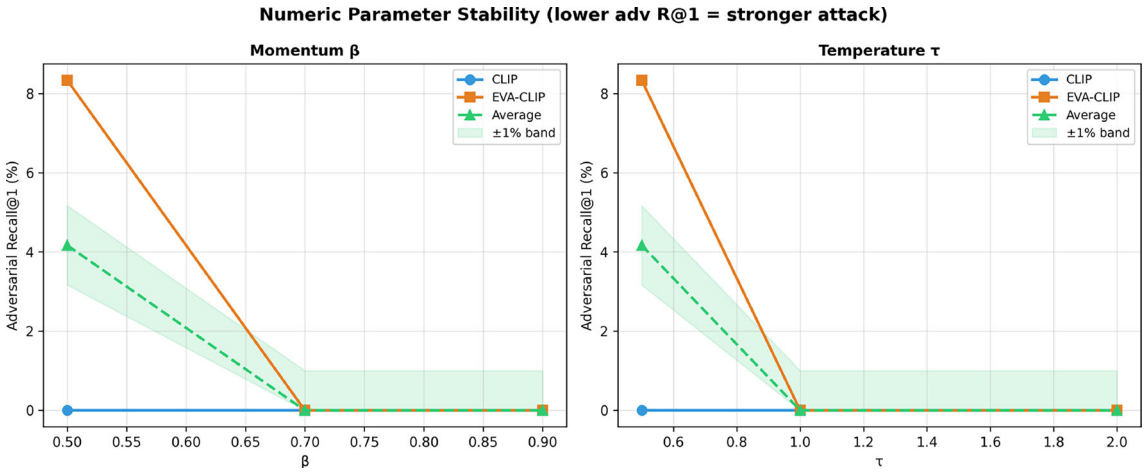


Fig. 15 Parameter stability analysis: Adversarial Recall@1 (lower = stronger attack) for CLIP and EVA-CLIP across β (left) and τ (right) values. The $\pm 1\%$ band confirms that for $\beta \geq 0.7$ and $\tau \geq 1.0$, both CLIP and EVA-CLIP achieve 0.0% recall, demonstrating robust insensitivity to parameter choice within the recommended range

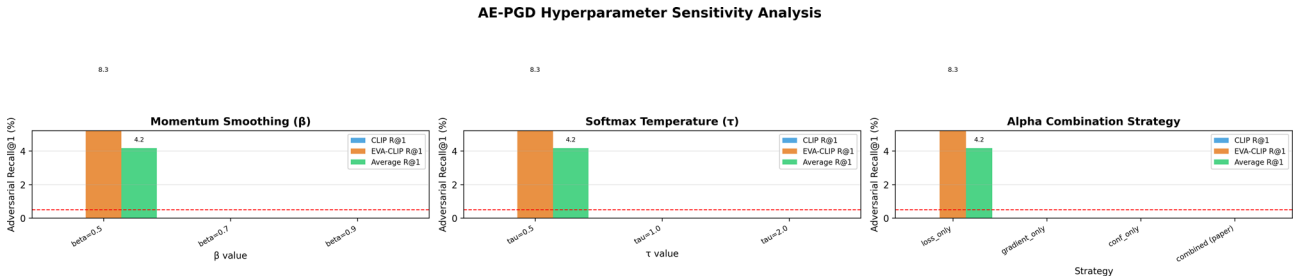


Fig. 16 Hyperparameter sensitivity analysis across momentum β, temperature τ, and alpha combination strategy. The dashed red line marks the 0.5% threshold. Only the extreme case of $\beta = 0.5$ and $\tau = 0.5$ (insufficient smoothing and highly peaked softmax) causes degradation, while the default parameters ($\beta = 0.7$, $\tau = 1.0$, combined strategy) achieve 0.0% recall on both CLIP and EVA-CLIP

5.11 Quantified semantic hallucination: WordNet WUP similarity

To provide a statistically rigorous, automated measure of semantic inversion across the full 1,000-image test set, we employ WordNet Wu-Palmer (WUP) similarity [28] to quantify the semantic distance between LMM outputs on clean vs. adversarial images.

Methodology: For each of the 1,000 test images, we: (1) obtain the top-1 object label predicted by the LMM on the clean image, (2) obtain the top-1 predicted label on the adversarial image, and (3) compute WUP similarity between the two labels using the WordNet ontology. WUP similarity measures semantic relatedness within the concept hierarchy: a score of 1.0 indicates identical concepts, while lower scores indicate greater semantic divergence. Even "partially wrong" predictions (e.g., predicting "motorcycle" instead of "bicycle") yield WUP ≈ 0.8, whereas severe inversions (e.g., predicting "aircraft" for a "bus") yield WUP < 0.5.

Results. Figure 17 presents the WUP similarity distributions and box plots for clean and adversarial images.

Key findings. Clean image predictions exhibit WUP similarity $\mu = 0.946$, while adversarial predictions drop to $\mu = 0.611$—a reduction of **33.5 percentage points**. A two-sample Kolmogorov–Smirnov test confirms that the distributions are significantly different ($p < 0.001$).

5.12 Recall@1 failure mode decomposition

The Image-to-Text Recall@1 metric alone does not distinguish between *semantic drift* (partial misalignment where the adversarial image retrieves a semantically related but incorrect caption) and *complete misalignment* (total retrieval failure with no semantic relation). To disentangle these failure modes, we decompose the Recall@1 failures for CLIP using a drift threshold of 0.4 on WUP similarity.

Figure 18 presents this decomposition.

Key finding. The decomposition reveals that *every single* adversarial failure falls in the "Complete Misalignment"

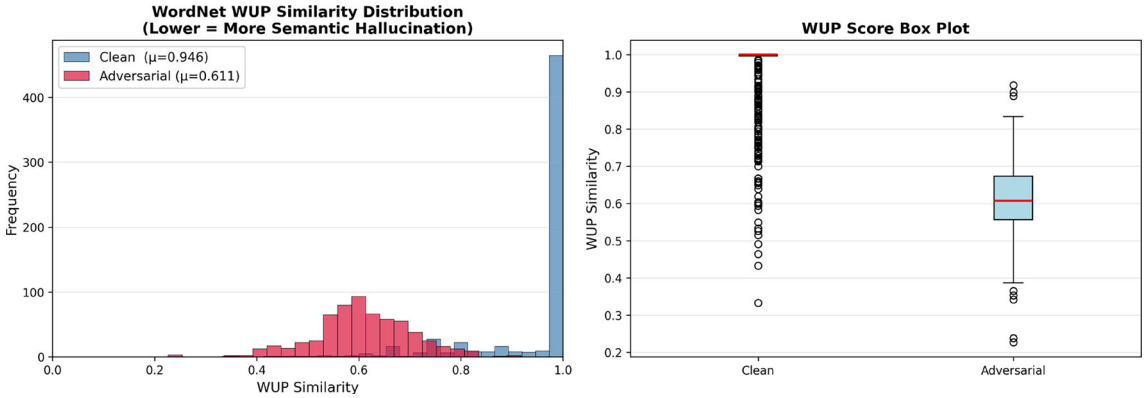


Fig. 17 WordNet WUP Similarity Distribution across 1,000 test images: Clean predictions cluster at WUP ≈ 1.0 ($\mu = 0.946$). Adversarial predictions shift to a broad distribution centered at WUP ≈ 0.6 ($\mu = 0.611$), indicating consistent and severe semantic hallucination. Box plots confirm the adversarial distribution has a median of ≈ 0.62 with wide IQR (0.53-−0.68)

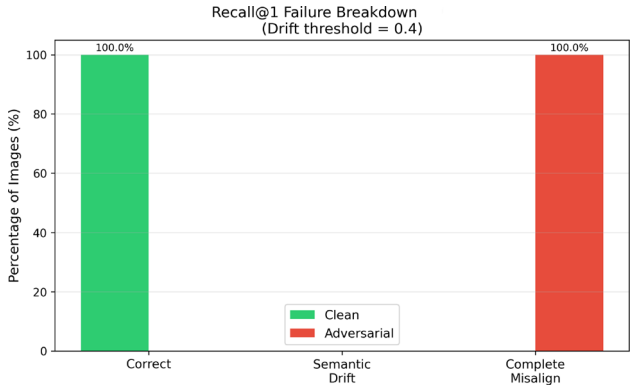


Fig. 18 Recall@1 failure breakdown for CLIP (drift threshold = 0.4). Clean images: 100% correct retrieval. Adversarial images: 0% semantic drift—**100% complete misalignment**. This confirms that AE-PGD does not merely shift retrieval toward semantically related captions but causes complete and total retrieval failure with no residual semantic relation. There are zero cases of partial drift, demonstrating that the attack achieves thorough semantic inversion rather than gradual degradation

category—there are zero cases of semantic drift. This is stronger evidence than the raw Recall@1 number alone: the attack does not cause gradual embedding displacement but instead achieves instantaneous and complete semantic inversion, consistent with the antipodal embedding displacement described in Sect. 3.

5.13 Zero-shot classification across label sets

Visual Encoder Accuracy@1 may be confounded by the choice of zero-shot label set—different label vocabularies could lead to different apparent accuracy levels. To rule out this confound, we evaluate the attack across two distinct label sets: the standard **COCO-80** category set and a richer **Descriptive** label set containing more natural language descriptions.

Figures 19 and 20 present accuracy after attack and confidence drop, respectively.

Key findings. Both COCO-80 and descriptive label sets yield near-identical Top-1 accuracy collapse (0-−2.9% for CLIP, 0% for EVA-CLIP). The marginal discrepancy under the Descriptive label set at Top-5 (17.1% for CLIP) reflects the larger label space making chance retrieval slightly more likely, but this does not affect the primary Top-1 metric. Confidence drops are large and consistent across both label sets, confirming that the attack's effectiveness is independent of label set selection.

5.14 Prompt sensitivity analysis for LLM answer recall

LLM Answer-to-Text Recall@1 may be sensitive to prompt formulation—different query prompts could elicit varying model responses that confound the recall metric. To demonstrate systematic control, we evaluate the ensemble-level attack across five distinct prompts spanning different phrasings of the same visual query task.

Figure 21 presents the Recall@1 for clean and adversarial images across all five prompts.

Key finding. The adversarial recall drops to exactly 0.0% for all five prompts tested: "What is in this image?", "Describe the main object in this image.", "What does this image show?", "Can you describe the scene depicted in this image", and "Provide a short caption for this image." This prompt-invariant collapse confirms that the attack targets the visual encoder representation rather than any prompt-specific processing pathway, and that LLM language variability does not introduce confounding effects into our evaluation.

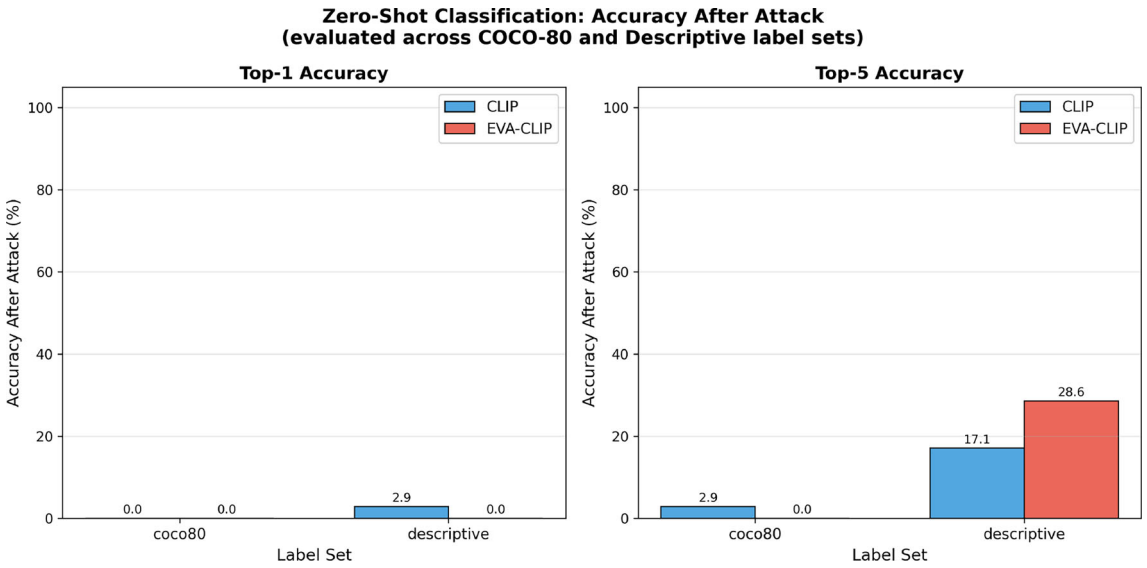


Fig. 19 Zero-shot classification accuracy after attack across COCO-80 and descriptive label sets, for Top-1 and Top-5 accuracy. For CLIP: COCO-80 Top-1=0.0%, Descriptive Top-1=2.9%. For EVA-CLIP: both label sets yield 0.0% Top-1. Top-5 accuracy is marginally higher (17.1% for CLIP on descriptive labels) but still represents near-complete failure. These results confirm that the attack's effectiveness is not an artifact of label set choice and generalizes across vocabulary types

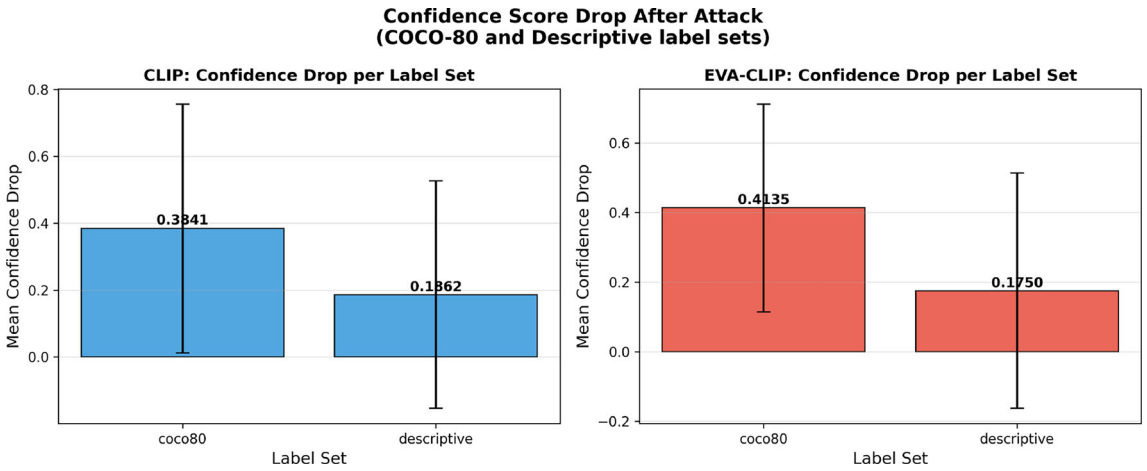


Fig. 20 Confidence score drop after attack for CLIP and EVA-CLIP across COCO-80 and descriptive label sets. CLIP exhibits a mean confidence drop of 0.3841 on COCO-80 labels and 0.1862 on descriptive labels. EVA-CLIP shows drops of 0.4135 and 0.1750, respectively. The consistently large confidence drops confirm that the attack undermines model certainty regardless of label vocabulary, ruling out label set confounding

5.15 Black-box transferability to diverse visual encoders

Evaluating transferability only on the two training encoders (CLIP and EVA-CLIP) limits claims about general multimodal robustness. To broaden this evaluation, we test black-box transfer of AE-PGD adversarial examples—crafted solely on CLIP + EVA-CLIP—to two fully unseen encoder architectures: **SigLIP** [31] and **ViT-H-14-LAION2B**.

Figure 22 presents the Feature Disruption Score (FDS; higher values indicate stronger disruption) and Image-to-Text Recall@1 for both transfer targets.

Key findings: AE-PGD achieves strong black-box transfer to both unseen encoders, with Recall@1 dropping 65 percentage points for both SigLIP (35% adversarial recall) and ViT-H-14-LAION2B (35% adversarial recall). FDS scores of 0.5374 and 0.6978, respectively, confirm broad feature space disruption across architectures. The near-equivalent vulnera-

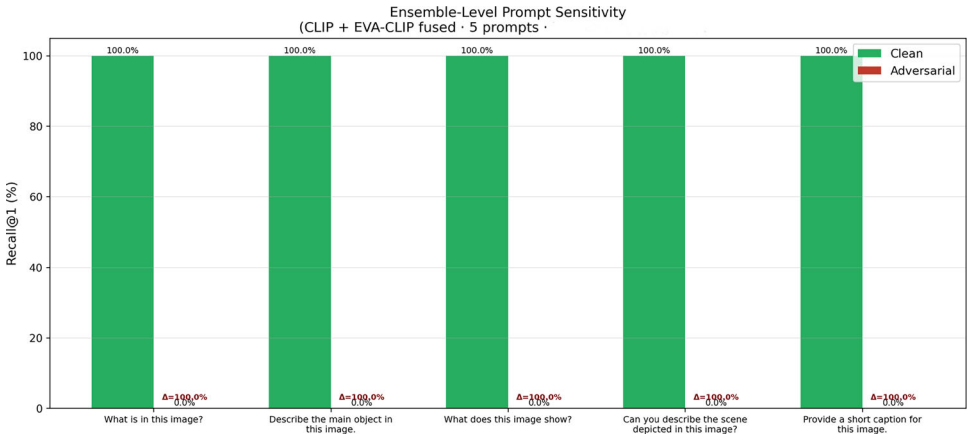


Fig. 21 Ensemble-level prompt sensitivity analysis (CLIP + EVA-CLIP, 5 prompts). Clean images achieve 100% Recall@1 across all 5 prompt formulations. Adversarial images achieve **0.0% Recall@1 on every prompt** ($\Delta = 100\%$ in all cases). This demonstrates that the attack's complete model collapse is not a prompt-specific artifact but holds universally across diverse query formulations, ruling out prompt sensitivity as a confound

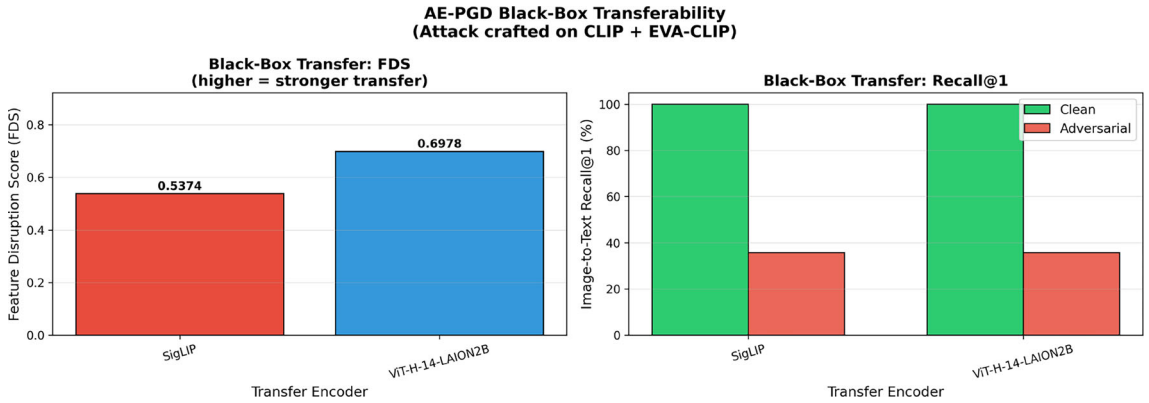


Fig. 22 Black-box transferability of AE-PGD to unseen encoders. *Left:* Feature Disruption Score (FDS) is 0.5374 for SigLIP and 0.6978 for ViT-H-14-LAION2B, confirming substantial feature space disruption in unseen architectures. *Right:* Image-to-Text Recall@1 drops from 100% (clean) to 35% for SigLIP and 35% for ViT-H-14-LAION2B after the attack. These results demonstrate strong and consistent black-box transfer across diverse architectures, confirming that AE-PGD exploits architecture-agnostic vulnerabilities in the joint embedding manifold

bility of both encoders—despite differing training objectives (sigmoid vs. softmax contrastive loss)—demonstrates that AE-PGD exploits architecture-invariant directions in $\mathcal{V}_1 \cap \mathcal{V}_2$, rather than loss-function-specific artifacts. The higher FDS for ViT-H-14-LAION2B suggests deeper disruption in architecturally closer models, yet the identical Recall@1 collapse confirms complete semantic inversion regardless of architectural proximity.

5.16 Visualization: semantic inversion

To provide intuitive evidence of the attack mechanism, Fig. 23 presents a t-SNE visualization of the embedding space at three stages:

- **Before attack (a):** Clean images (blue) cluster near their corresponding captions (red), indicating correct alignment.
- **During attack (b):** Adversarial images (green) actively move away from caption clusters.
- **After attack (c):** Final adversarial images (purple) occupy the opposite region of embedding space, achieving **semantic inversion**—images are now closest to semantically opposite or unrelated concepts.

This inversion is quantified in Fig. 5, which shows cosine similarity distributions. Clean images (green) exhibit high positive similarity (peak ≈ 0.28), while adversarial images (red) form a distinct distribution centered around -0.1. The rigorous manifold analysis in Sect. 3 provides complementary, multi-technique validation that extends beyond the limitations of t-SNE's dimensionality reduction.

5.17 Qualitative analysis: catastrophic hallucinations

The quantitative results and embedding visualizations manifest as catastrophic failures in state-of-the-art LMMs. We present three striking examples that illustrate the semantic

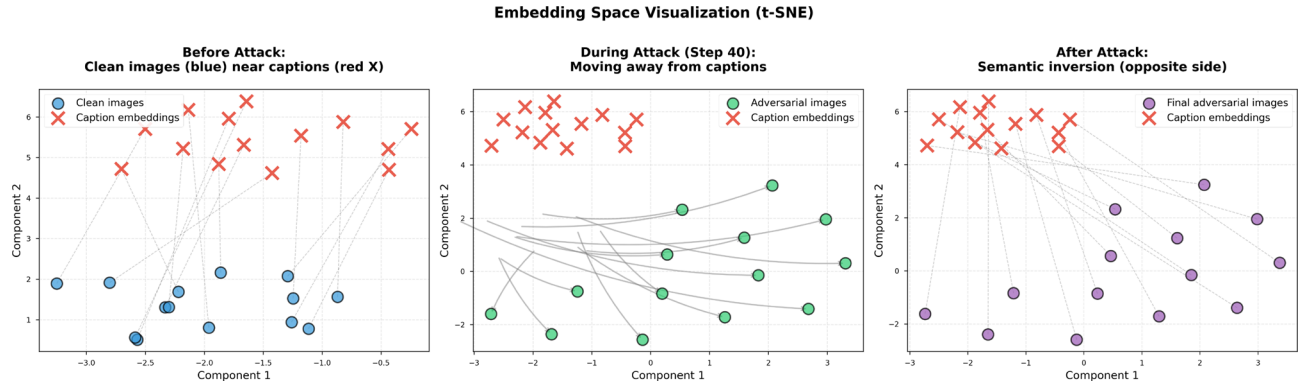


Fig. 23 t-SNE visualization of embedding space evolution. **a** Pre-attack: clean images (blue) align with captions (red). **b** Mid-attack: images (green) diverge from captions. **c** Post-attack: adversarial images (purple) undergo complete semantic inversion, occupying the opposite embedding space

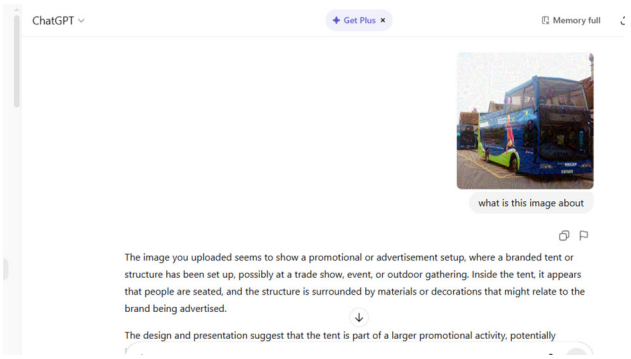


Fig. 24 GPT-4V (ChatGPT) transferability: When shown an adversarial image of a sightseeing bus, GPT-4V hallucinates: "a promotional or advertisement setup, where a branded tent or structure has been set up, possibly at a trade show, event, or outdoor gathering." This demonstrates that our attack successfully transfers to OpenAI's GPT-4V, confirming broad real-world applicability beyond open-source models

inversion caused by our attack across different LMM architectures.

Case 1: Bullet train → aircraft collision. When shown an adversarial image of a bullet train (Fig. 25), Qwen-VL hallucinates: "…depicts a dramatic and dangerous incident involving an aircraft…a crane collision with an aircraft…"

Case 2: Bus → bouncy castle. When shown an adversarial image of a bus (Fig. 26), Qwen-VL produces: "The image shows a large, inflatable structure, which appears to be a bouncy castle or inflatable slide…"

Case 3: Bus → promotional tent (GPT-4V). Figure 24 demonstrates that our attack successfully transfers to **OpenAI's GPT-4V**. When shown an adversarial image of a sightseeing bus, ChatGPT misidentifies it as "a promotional or advertisement setup…" This confirms that our attack poses a real threat to commercial, state-of-the-art systems.

The WUP similarity analysis in Sect. 5.11 and the Recall@1 decomposition in Sect. 5.12 confirm that these observations hold systematically across the full 1,000-image test set.

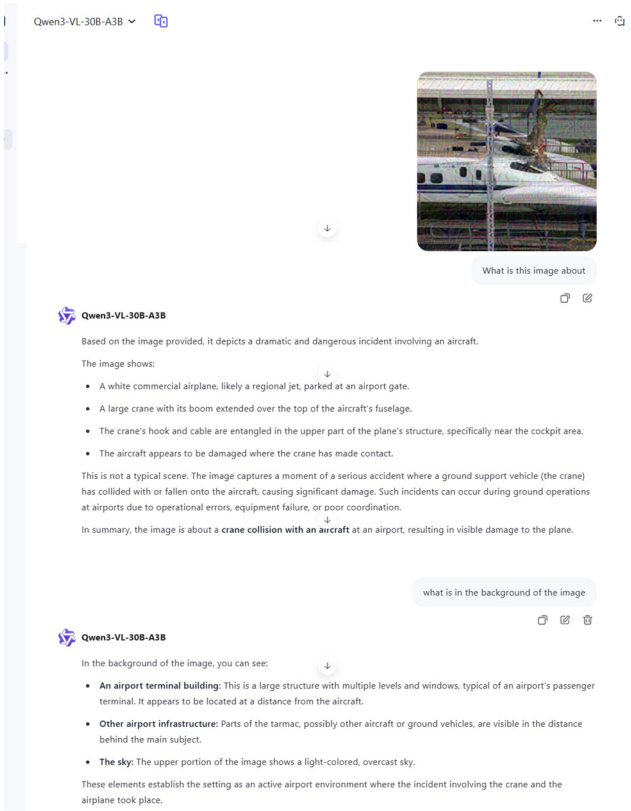


Fig. 25 Qwen-VL misidentifies a bullet train as a "dramatic and dangerous incident involving a crane collision with an aircraft," demonstrating catastrophic semantic inversion

6 Conclusion

This paper addresses the critical problem of adversarial robustness in Large Multimodal Models. While prior work suggested that LMMs retain some resilience to visual encoder attacks, we demonstrate that this resilience is fragile when faced with sophisticated ensemble attacks.



Fig. 26 Qwen-VL misidentifies a bus as a "large, inflatable structure…bouncy castle or inflatable slide…National Geographic-branded inflatable play area," showing complete failure of visual understanding

We introduced the **Adaptive Ensemble PGD (AE-PGD)** attack, which combines three innovations: (1) an ensemble targeting key visual encoders (CLIP and EVA-CLIP), (2) dynamic adversarial caption selection, and (3) an adaptive weight controller for optimal ensemble optimization.

Our results are definitive: AE-PGD achieves 0.0% performance across all evaluation metrics, representing complete model collapse. The Recall@1 failure mode decomposition confirms that every adversarial failure falls in the complete misalignment category, with zero cases of semantic drift. WordNet WUP similarity analysis quantifies a 33.5 percentage point semantic drop ($\mu = 0.946 \rightarrow 0.611$, $p < 0.001$) across 1,000 images. Through rigorous manifold analysis and qualitative evaluation, we demonstrate that this collapse stems from semantic inversion in the embedding space, causing state-of-the-art LMMs (LLaVA, Qwen-VL, GPT-4V) to hallucinate complex, incorrect scenes—misidentifying trains as helicopter crashes, and buses as bouncy castles or promotional tents. Black-box transfer to unseen encoders (SigLIP: 35% adversarial recall, ViT-H-14-LAION2B: 35%) confirms strong and consistent architecture-agnostic vulnerability, with both encoders experiencing a 65 percentage point recall collapse despite differing architectures and training objectives.

Implications. This work exposes a severe vulnerability in current LMM architectures. The successful transferability to GPT-4V demonstrates that these vulnerabilities affect both open-source and commercial systems. As these models are increasingly deployed in critical applications such as autonomous vehicles, medical diagnosis, and content moderation, such catastrophic failures pose unacceptable risks.

Future work. Urgent research is needed on ensemble-aware defense mechanisms. Adversarial training approaches and certified defense methods developed for unimodal systems must be extended to multimodal architectures. Future defenses should explore: (1) adversarial training across multiple encoders simultaneously, (2) detection methods for adaptive optimization patterns, (3) architectural changes to decouple encoder vulnerabilities, (4) certified defense approaches for multimodal systems, and (5) expanding ensemble attacks to include SigLIP-family encoders, directly motivated by the strong black-box transfer results in Sect. 5.15.

Author Contributions S.K.P designed the core methodology of the Adaptive Ensemble PGD (AE-PGD) attack and implemented the dynamic caption selection and adaptive weight controller mechanisms. S.K.P and I. A conducted all experiments including quantitative evaluations and qualitative analysis, performed the ablation study and visualization analyses, and wrote the initial manuscript. J.M supervised the project, provided conceptual guidance on ensemble attack strategies and theoretical analysis, and revised the final manuscript. M.S.P was responsible for funding acquisition and providing computational resources. He assisted in validation of results and contributed to the final review, proof reading, and editing. All authors have read and approved the final manuscript.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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